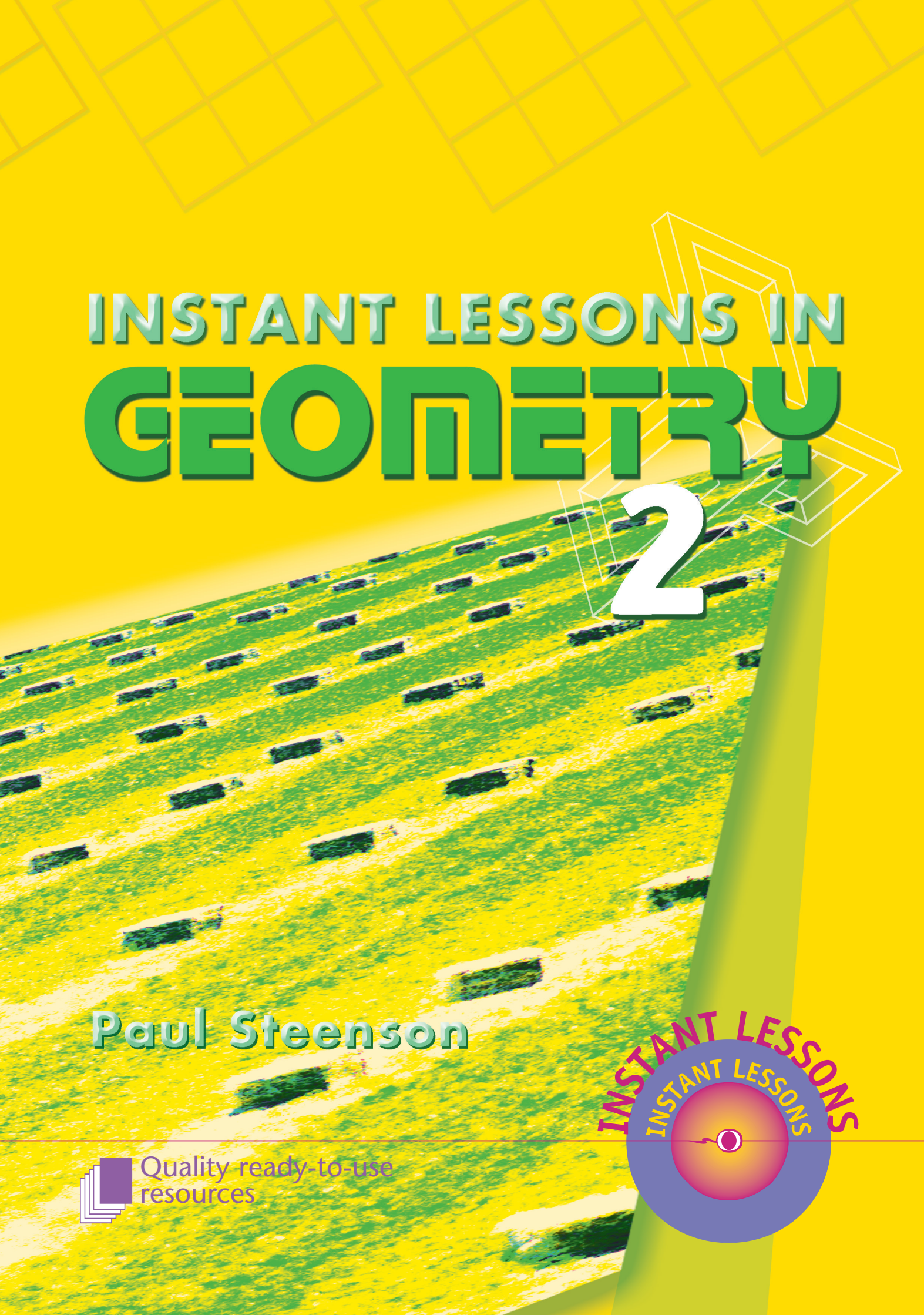
INSTANT LESSONS IN GEOMETRY 2



Paul Steenson

INSTANT LESSONS

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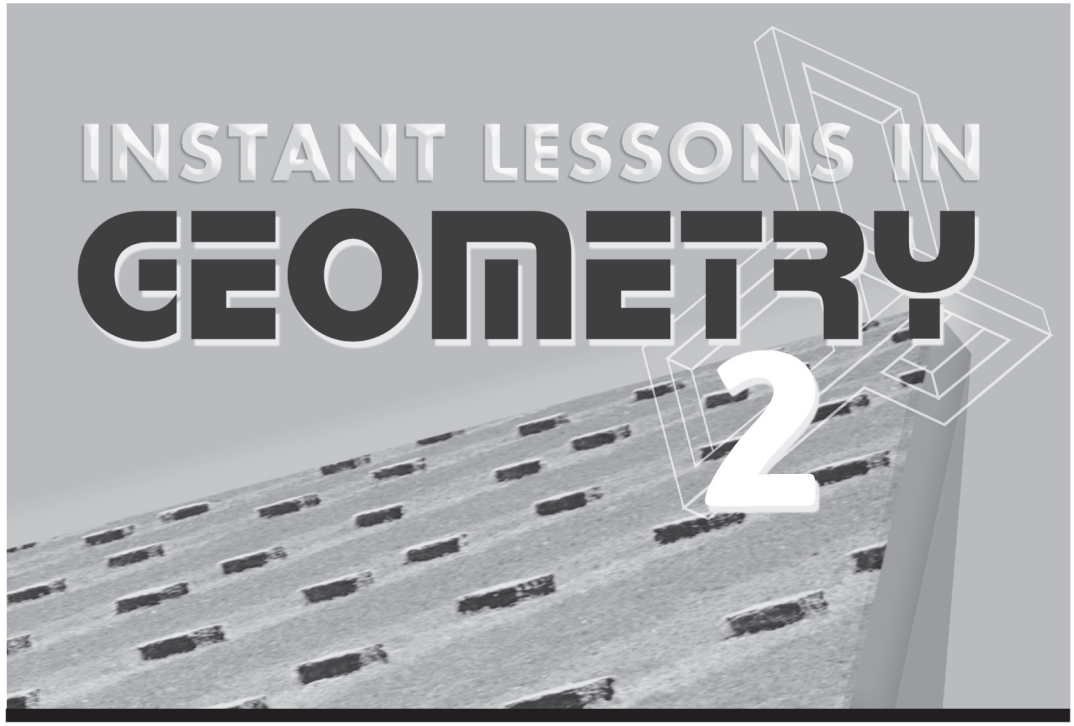

Paul Steenson

INSTANT LESSONS

Paul Steenson has more than thirty years' experience teaching Mathematics in secondary schools in Australia. He has been Head of Department since 1984 in a variety of schools in NSW. He has also worked as a test development officer and been an assessor of Mathematics papers for the NSW HSC examination. Paul served as a member of the NSW Mathematics Syllabus Committee and has been a marker of both School Certificate and HSC examination papers. He has been a regular presenter of curriculum workshops and teacher development activities. He has been a co-author of textbooks and written a number of titles in the *Instant Lessons* series.

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Commissioning editor: Jennifer Gough
Managing editor: Ruth Schultz
Typesetting: Thomson Digital, Final Page Editing & Design
Editor: Ruth Schultz
Cover design: Gail McManus Graphics
Printed in Australia by Ligare Book Printer

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ISBN 978 0 99542 759 4

Introduction

This book provides a systematic and comprehensive coverage of selected topics from the junior mathematics curriculum. It is independent of any textbook and is not linked to any syllabus. Further geometry topics can be found in *Instant Lessons in Geometry Book 1*.

Most worksheets follow a format comprising three sections:

- **Section A** consists of a set of introductory questions or review of previous knowledge, which can be used as a lesson starter or 'warm up'.
- **Section B** provides an investigation, skills development or drill, with graded questions in the focus area of the worksheet.
- **Section C** provides further practice in the form of different types of questions, challenge or extension questions.

Review worksheets are also included in each topic area.

There are minimal examples or explanations as teachers have diferent methods of working that they would like their students to follow. Most worksheets could be used by students in a range of year levels or courses. The teacher may decide to use only one or two sections from a worksheet in a lesson, or as a homework assignment. The worksheets are designed to be flexible to meet the needs of the teacher and students.

A time required for students to complete each worksheet is not suggested as this will vary significantly depending on the year level of the students and their level of mathematical skill. The teacher is the best person to make this judgement.

Suggested answers have been provided for most questions, including some of the guided proofs. In general, answers have not been provided for practical tasks, constructions, sketch diagrams, proofs or literacy activities.

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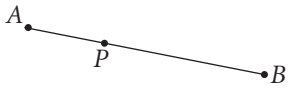
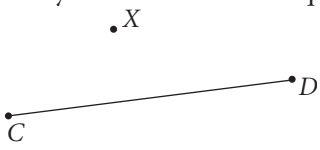
Geometric constructions 1.1

A

1 Draw a line *AB*. Mark a point *P* on this line. Use a set square to draw a line through *P*, perpendicular to *AB*.



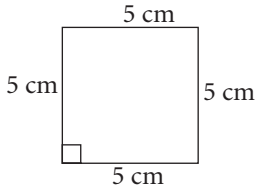
2 Draw a line *CD* and mark a point *X*, not on this line. Use your ruler and a set square to draw a line:



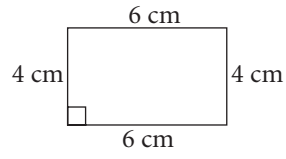
a through *X*, perpendicular to *CD*

b through *X*, parallel to *CD*.

3 Use a ruler and a set square to construct a square with sides 5 cm long. (Remember that all angles are right angles.)



4 Use a ruler and a set square to construct a rectangle with sides measuring 6 cm and 4 cm.



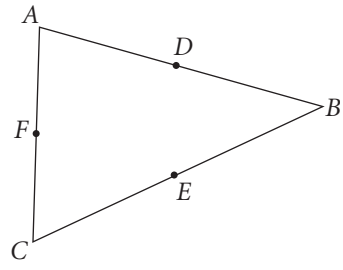
5 Draw any triangle *ABC*.

a Use your ruler to mark the midpoint (half-way point) of the sides *AB*, *BC* and *CA*.

b Label these points *D*, *E* and *F* as shown.

c Use your ruler to join *A* to *E*, *B* to *F* and *C* to *D*.

d What do you notice about the lines *AE*, *BF* and *CD*?



B Bisecting an interval

1 Using compasses and a ruler, follow the instructions to construct the perpendicular bisector of an interval.

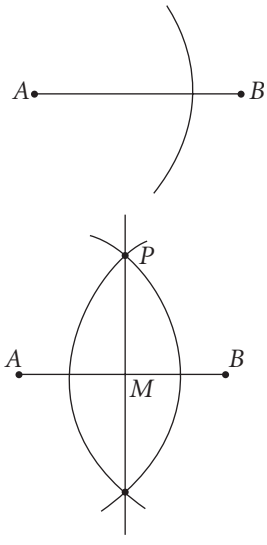
a Draw an interval *AB* on your working paper. (Leave space on each side to complete this construction.)

b Open your compasses to more than half the length of *AB*. Place the point of your compasses on *A* and drawn an arc.

c Using the same radius, place the compass point on *B*. Draw a second arc.

d Join the points where the two arcs meet, and label points *P* and *M* as shown. This line is the perpendicular bisector of the interval *AB*.

e Check by measurement that $AM = MB$ and $\angle AMP = 90°$.



2 **a** Draw an interval *PQ*.

b Use the construction method in question 1 to construct the perpendicular bisector of the interval *PQ*.

c Check the accuracy of your construction by measurement.

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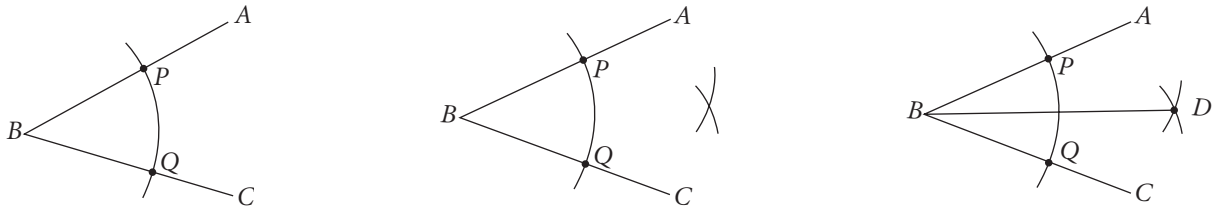
Bisecting an angle

3 Using compasses and a ruler, follow the instructions to bisect an angle.



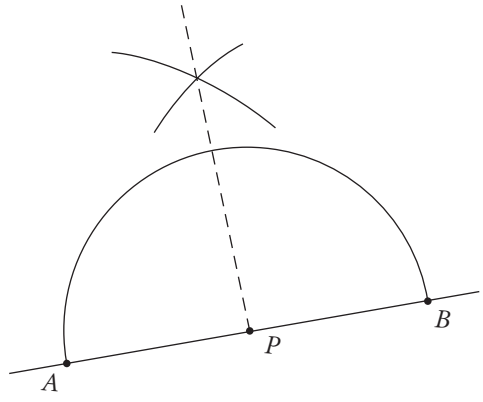
- **a** Use your pencil and ruler to draw any angle *ABC* on your working paper.
- **b** Place the point of your compasses on the vertex of the angle (point *B*) and draw an arc that cuts the arms of the angle at points *P* and *Q*.
- **c** Place the compass point on *P* and draw an arc inside the angle.
- **d** Using the same radius, place the compass point on *Q* and draw a second arc, as shown.
- **e** Label the point *D*, where the arcs meet. Join *B* and *D*. The line *BD* bisects the angle *ABC*.
- **f** Use a protractor to check whether your construction was accurate.

4
- **a** Draw an angle *LMN*.
- **b** Use the construction method in question 3 to bisect this angle, using compasses and a ruler.
- **c** Use a protractor to check whether your construction was accurate.

C Construct a perpendicular to a line

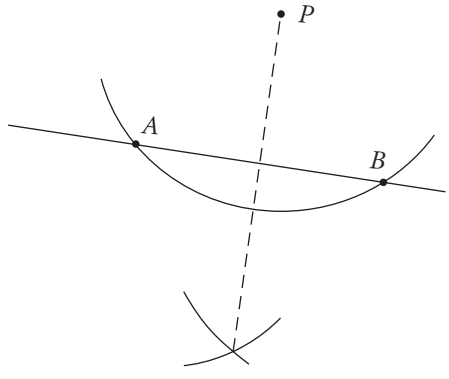
1 Use this construction to draw a perpendicular to a line through a given point on the line.

- **a** Draw a line and mark a point *P* on the line.
- **b** Using *P* as the centre, draw half a circle, as shown. Label points *A* and *B*.
- **c** Increase the radius on your compasses.
- **d** Place the point of your compasses on *A* and draw an arc above the point *P*.
- **e** Using the same radius, place the compass point on *B* and draw a second arc above the point *P*.
- **f** Join the point where the two arcs meet to *P*.
- **g** Use a set square to check that the two lines meet at right angles.



2 Use this construction to draw a perpendicular to a line, through a given point *not* on the line.

- **a** Draw a line and mark a point *P*, not on the line.
- **b** Place the point of your compasses on *P* and draw an arc that cuts the line in two places, *A* and *B*.
- **c** Place the point of your compasses on *A* and draw an arc on the opposite side of the line from *P*.
- **d** Using the same radius, place the compass point on *B* and draw a second arc on the opposite side of the line from *P*.
- **e** Join the point where the two arcs meet to *P*.
- **f** Use a set square to check that the two lines meet at right angles.



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Constructing triangles 2.1

A

To construct a triangle congruent to a given figure, we need to know three measurements (lengths of sides or the size of angles). This information needs to follow the pattern of the tests for congruent triangles:

- The lengths of the 3 sides (SSS test)
- The lengths of 2 sides and the size of the angle between them (SAS test)
- The sizes of 2 angles and the length of 1 side (AAS test)

Construct a triangle given the lengths of the 3 sides.

1 Construct a triangle *ABC* with sides 6 cm, 4 cm and 5 cm. Follow these steps:

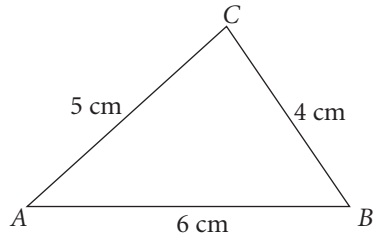
a Using a ruler, draw an interval *AB* that is 6 cm long.

b Using compasses, draw an arc with radius 4 cm, centred at *B*.

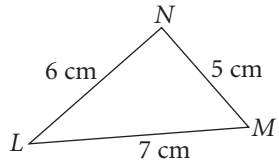
c Now draw a second arc with radius 5 cm, centred at *A*.

d Where these arcs cross, mark the point *C*, the third vertex of the triangle.

e Use a ruler to draw the sides *AC* and *BC* to complete the triangle.



2 Use a ruler and a pair of compasses to construct a triangle *LMN* with sides 7 cm, 5 cm and 6 cm.



Construct a triangle given the lengths of 2 sides and the size of the angle between them.

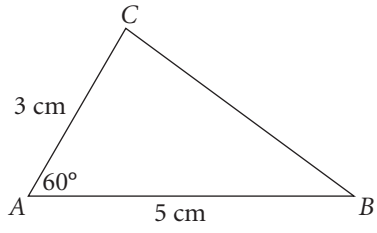
3 Construct a triangle *ABC* with sides 5 cm and 3 cm and the angle between them measuring 60°. Follow these steps:

a Using a ruler, draw an interval *AB* that is 5 cm long.

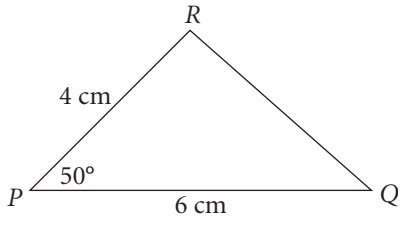
b Use a protractor to construct $\angle CAB$ measuring 60°.

c Use a ruler to mark the point *C*, 3 cm from *A*.

d Use a ruler to draw the side *BC*, completing the triangle.



4 Use a ruler and a protractor to construct a triangle *PQR* in which $PQ = 6$ cm, $PR = 4$ cm and $\angle RPQ = 50°$.



Construct a triangle given the sizes of 2 angles and the length of 1 side.

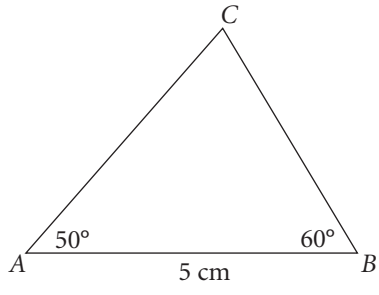
5 Construct a triangle *ABC* with $\angle A = 50°$, $\angle B = 60°$ and the side $AB = 5$ cm. Follow these steps:

a Using a ruler, draw an interval *AB* that is 5 cm long.

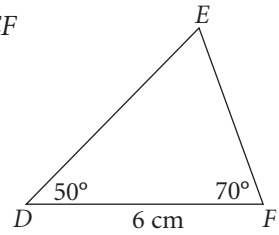
b Use a protractor to construct $\angle A$ measuring 50°.

c Now construct $\angle B$ measuring 60°.

d Mark the point *C* where these two lines meet and complete the triangle.



6 Use a ruler and a protractor to construct a triangle *DEF* with $\angle D = 50°$, $\angle F = 70°$ and the side $DF = 6$ cm.



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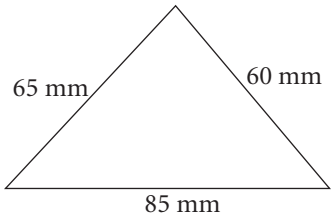
B

Use a ruler and a protractor or a pair of compasses to construct these triangles.

1



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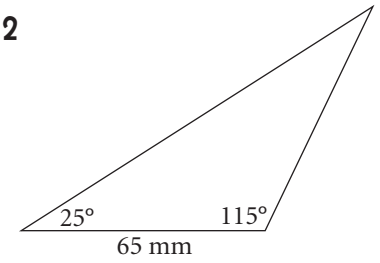

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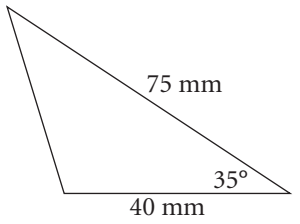


4 Triangle PQR with sides $PQ = 5$ cm, $PR = 5$ cm and $QR = 3$ cm

5 Triangle XYZ with sides $XY = 65$ mm, $XZ = 55$ mm and $YZ = 35$ mm

6 Triangle DEF with $\angle D = 35°$, $\angle F = 75°$ and $DF = 8$ cm

7 Triangle RST with $\angle S = 40°$, $\angle T = 65°$ and $ST = 75$ mm

8 Triangle LMN in which $MN = 7$ cm, $ML = 5$ cm and $\angle LMN = 70°$

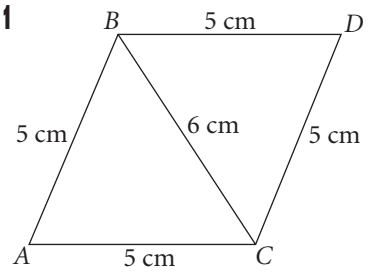
9 Triangle ABC in which $AC = 85$ mm, $AB = 65$ mm and $\angle BAC = 55°$

C

Similar methods can be used to construct these quadrilaterals. For each quadrilateral:

- use compasses and a ruler to construct the triangle ABC
- construct triangle BDC using the side BC of ΔABC as your base.

1



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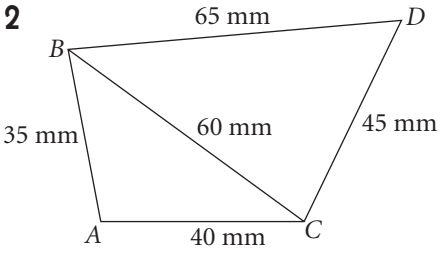

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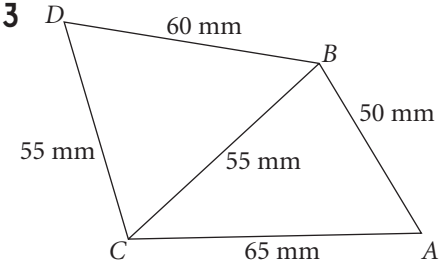

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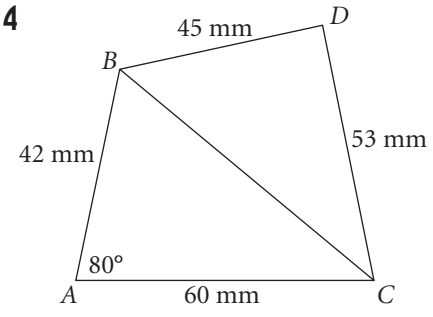

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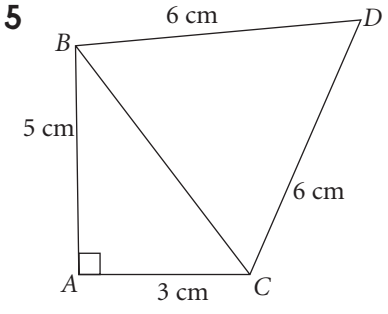

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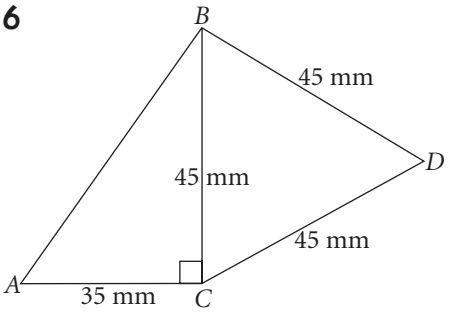

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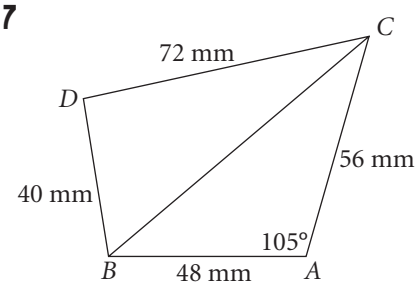

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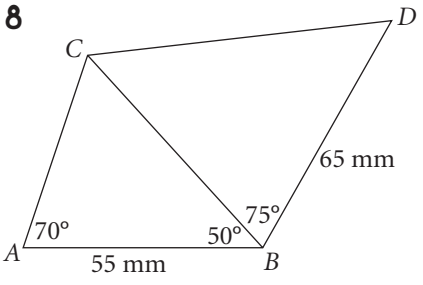

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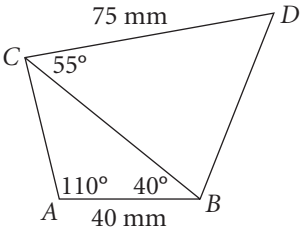

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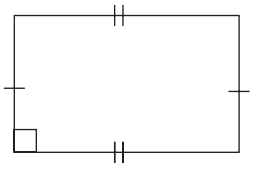
Constructing quadrilaterals 3.1

A

Quadrilaterals are plane figures bounded by four straight sides.
Types of quadrilateral: parallelogram, rectangle, square, rhombus, trapezium, kite.

1 State the type of quadrilateral in each diagram.

a



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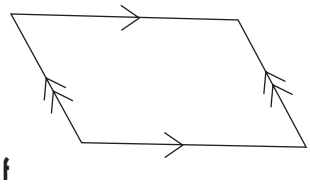

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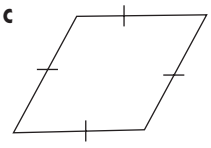

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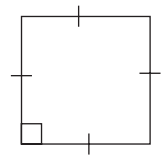

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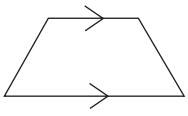

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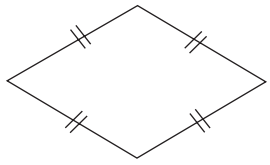

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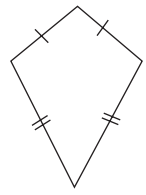

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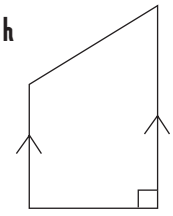
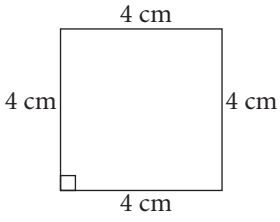
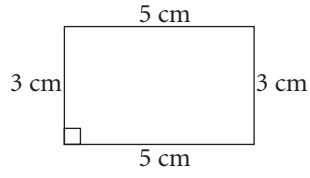


2 **a** Use a ruler and a set square to construct a square with sides 4 cm long. (Remember that all angles are right angles.)



b Construct a square with sides 6 cm long.

3 **a** Use a ruler and a set square to construct a rectangle with sides measuring 5 cm and 3 cm.



b Construct a rectangle with sides measuring 65 mm and 35 mm.

B

Use a ruler and a protractor to construct these quadrilaterals.

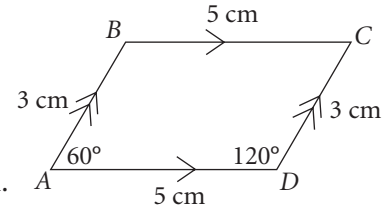
1 **a** Draw the line AD 5 cm long.

b Use your protractor to construct $\angle A$ measuring 60°.

c Now construct $\angle D$ measuring 120°.

d Use your ruler to mark the points B and C so that $AB = 3$ cm and $DC = 3$ cm.

e Use your ruler to join the points B and C, completing the parallelogram.



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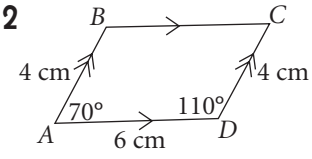

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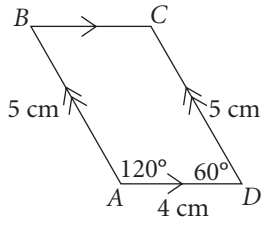

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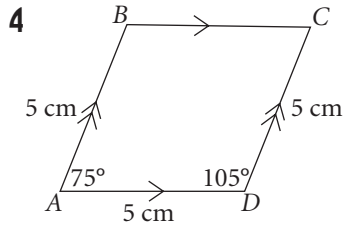

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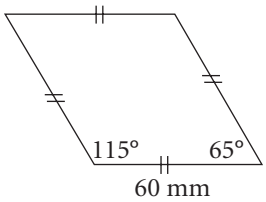

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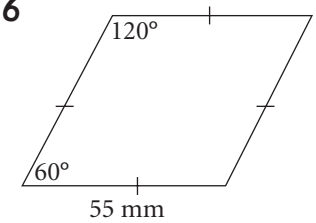

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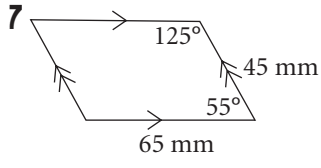

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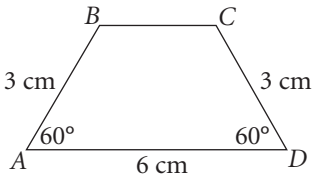
8 A parallelogram with sides 8 cm and 5 cm and one pair of opposite angles measuring 50°

9 A rhombus with sides measuring 65 mm and one pair of opposite angles measuring 60°

10



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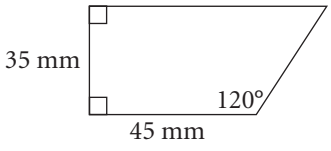

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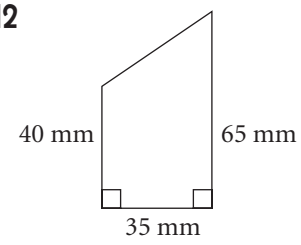

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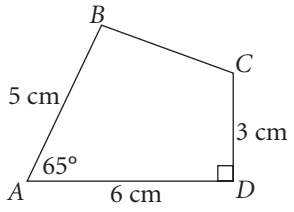

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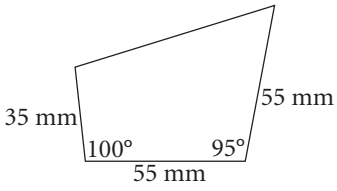

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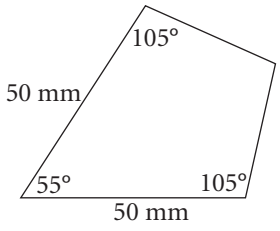

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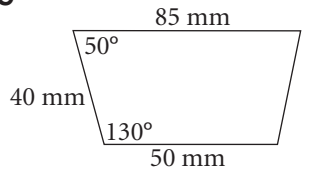


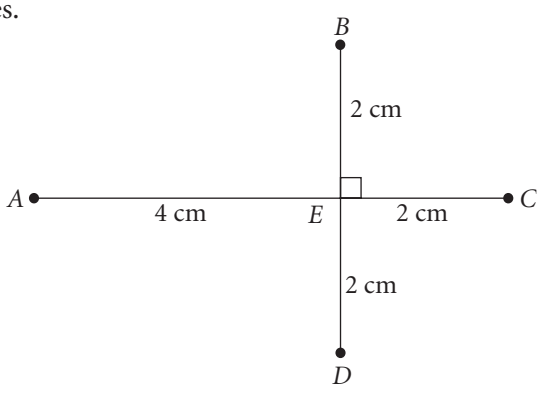
C The diagonals of a square, a rhombus and a kite meet at right angles. The following constructions are based on this property.

1 **a** Use a ruler and set square to construct the lines *AC* and *BD*, which intersect at right angles, as shown.

b Join the points *A*, *B*, *C* and *D* to form a quadrilateral.

c Measure the lengths of the sides of this quadrilateral (in millimetres).

d What type of quadrilateral is *ABCD*?

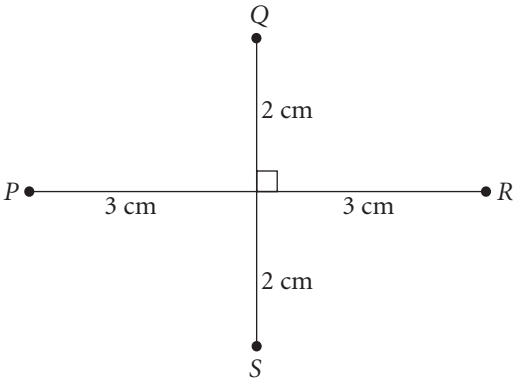


2 **a** Use a ruler and set square to construct the lines *PR* and *QS* which bisect each other at right angles.

b Join the points *PQRS* to form a quadrilateral.

c Measure the lengths of the sides of this quadrilateral (in millimetres).

d What type of quadrilateral is *PQRS*?



3 Construct a rhombus with diagonals measuring 7 cm and 4 cm. (Remember that the diagonals of a rhombus bisect each other at right angles.)

4 Use the method shown in question 2 to construct a square with diagonals measuring 5 cm. (Remember that the diagonals of a square are the same length.)

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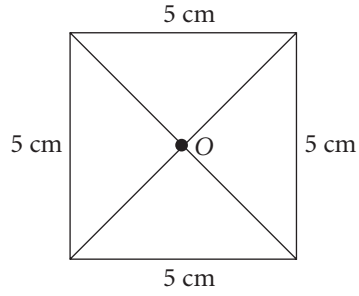
Constructions and circles 4.1

A



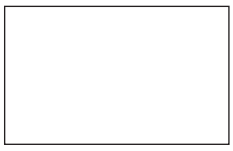
1 **a** Using a ruler and set square, construct a square with sides 5 cm long.

b Draw the two diagonals of this square. Label the point *O*, their point of intersection.

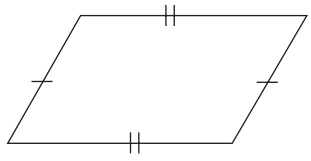
c Use compassses to draw a circle to pass through the four vertices of the square. (The point *O* is the centre of this circle.)

2 Investigate which of the following quadrilaterals can have a circle drawn through their four vertices. Draw the diagonals and follow the same steps you used in question 1.

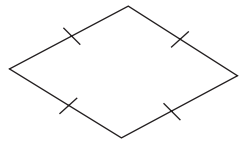
a rectangle



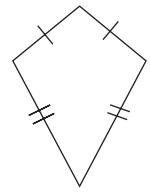
b parallelogram



c rhombus



d kite



3 Follow these steps to construct an *equilateral triangle* inside a circle.

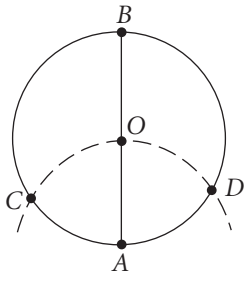
a Use a pair of compasses to construct a circle with radius 4 cm. Label the point *O*, the centre of the circle.

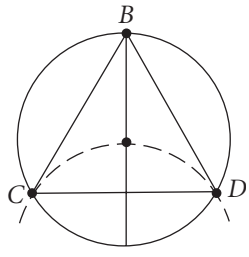
b Draw a diameter *AB*.

c Using *A* as the centre and *OA* as radius, draw an arc intersecting the circle at points *C* and *D*.

d Use your ruler to join the points *C*, *B* and *D*, forming a triangle.

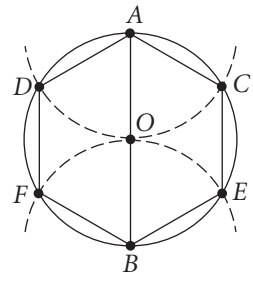
e Measure the lengths of the sides *BC*, *BD* and *CD* to confirm that triangle *BCD* is equilateral.





B

1 Follow these steps to construct a *regular hexagon* inside a circle.



a Use a pair of compasses to construct a circle with radius 4 cm (or more). Label the point *O*, the centre of the circle.

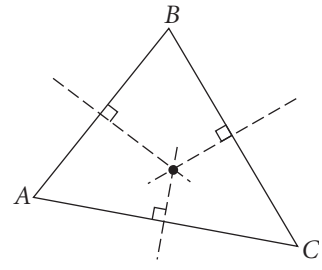
b Draw a diameter *AB*.

c Using *A* and *B* as centres and *OA* as radius, draw arcs intersecting the circle at points *C*, *D*, *E* and *F*.

d Use a ruler to join the points *A*, *C*, *E*, *B*, *F* and *D* to form the hexagon *ACEBFD*.

e Measure the lengths of each side of your hexagon to confirm that they are equal. (The six sides of a regular hexagon are the same length.)

2 **a** Draw any triangle *ABC*.



b Construct the perpendicular bisectors of each side. They should be concurrent (meet at the same point).

c Using this point of intersection as centre, draw a circle which passes through *A*, *B* and *C*, the vertices of the triangle. This is called the *circumcircle* of the triangle.

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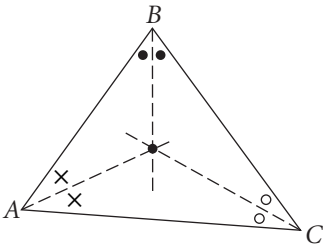
3 **a** Draw any triangle *ABC*.

b Bisect each angle of the triangle. These lines should be concurrent.

c Using the point of intersection of the bisectors as centre, draw a circle which just touches all three sides of the triangle.



4 Construct two circles with radii 3 cm and 4 cm that just touch at one point so that:

a they touch externally

b the smaller circle is inside the larger circle.

C

1 Construct an equilateral triangle with sides that are chords of a circle of radius 5 cm.

2 Mark two points *A* and *B*, 6 cm apart.

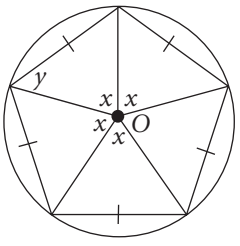
a Construct a circle with *AB* as diameter.

b Construct a square with *A* and *B* as opposite vertices.

3 The diagram shows a regular pentagon inscribed in a circle, centre *O*.

a What is the size of each angle *x*?

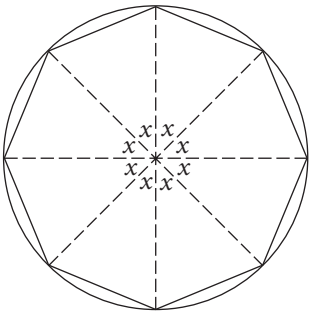
b What is the size of each angle *y*?



4 Construct a circle with radius 5 cm. Construct a regular pentagon with its vertices on the circumference of this circle.

5 **a** This diagram shows a regular octagon inscribed in a circle. The diagonals shown all pass through the centre of the circle. What is the size of each angle marked *x*?

b Construct a circle with radius 5 cm. Construct a regular octagon with its vertices on the circumference of this circle.



6 Check the meanings of the following mathematical terms. Write a sentence to explain the meaning of each term. (You could draw diagrams to help.)

a altitude of a triangle

b bisector of an angle

c centroid of a triangle

d circumcentre

e incentre

f incircle

g median of a triangle

h orthocentre

i perpendicular bisector

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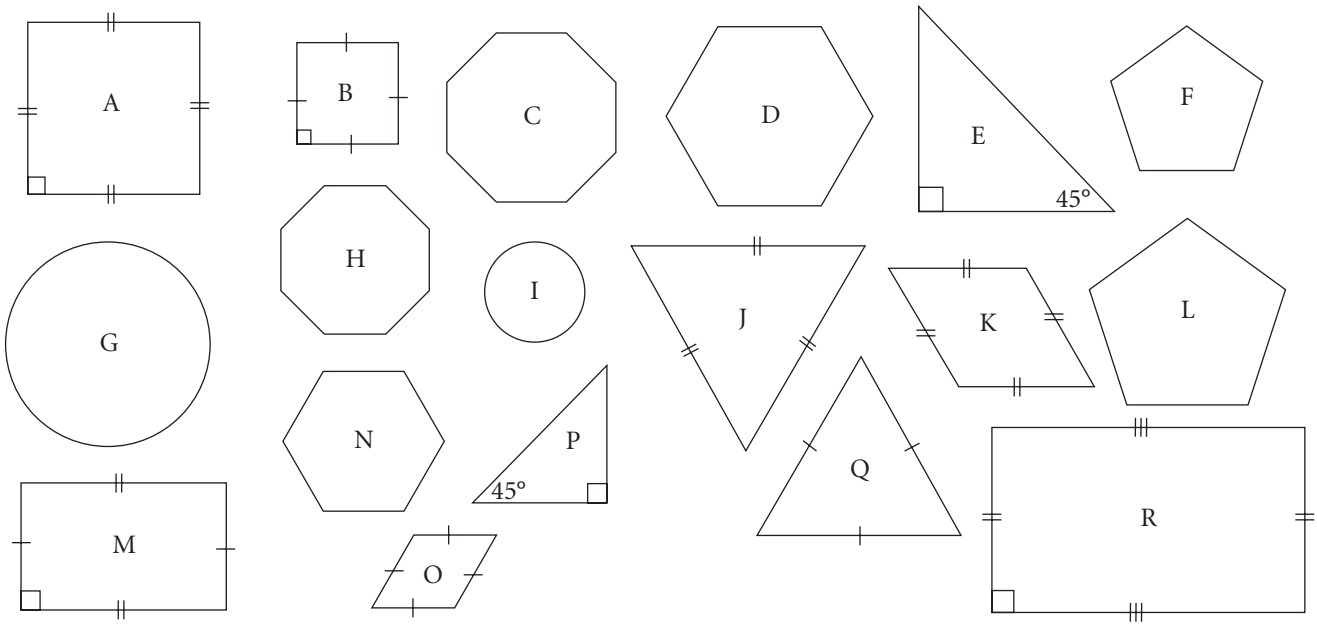
NOTE: DIAGRAMS NOT TO SCALE

Similar figures 5.1

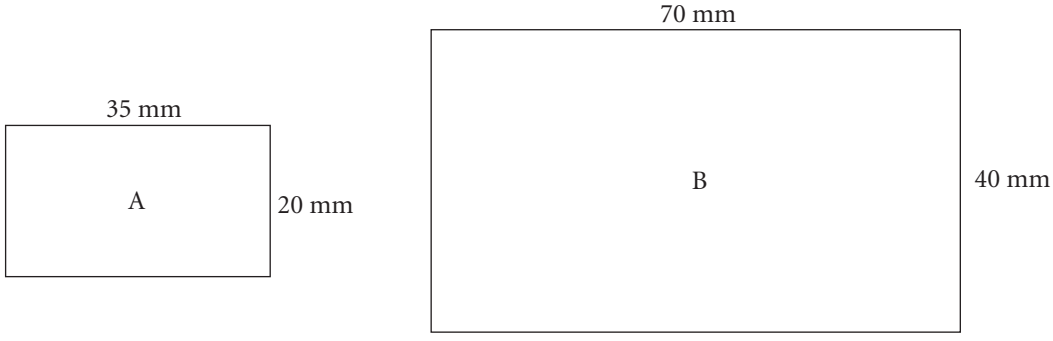
A

Similar figures have the same shape but are not the same size. Matching angles are the same size. Matching sides are in the same ratio. One figure is an enlargement of the other figure.

1 Match the pairs of similar figures.



2 Rectangle A has been enlarged to give rectangle B. They are similar.



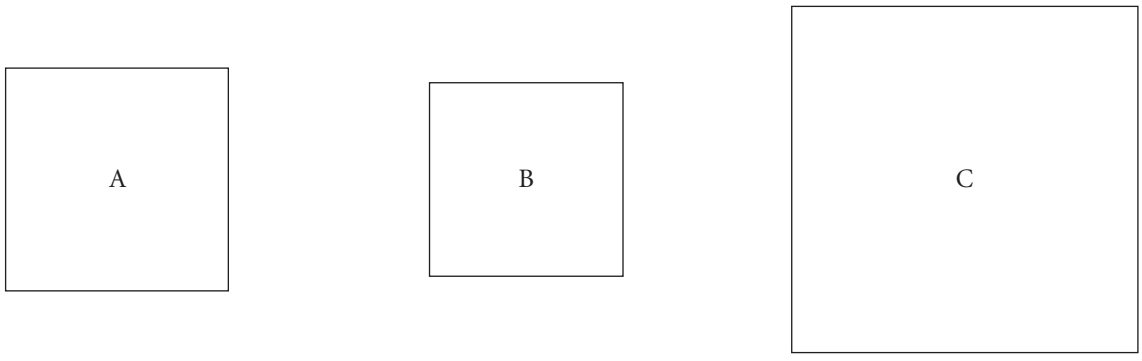
a Complete these statements:

The length of rectangle B is _____ times the length of rectangle A.

The breadth of rectangle B is _____ times the breadth of rectangle A.

b What is the ratio of the matching sides of these rectangles?

3 Measure the sides of these squares, to the nearest millimetre.



a Are all three squares similar?

b 'Any two squares are similar' – is this true or false?

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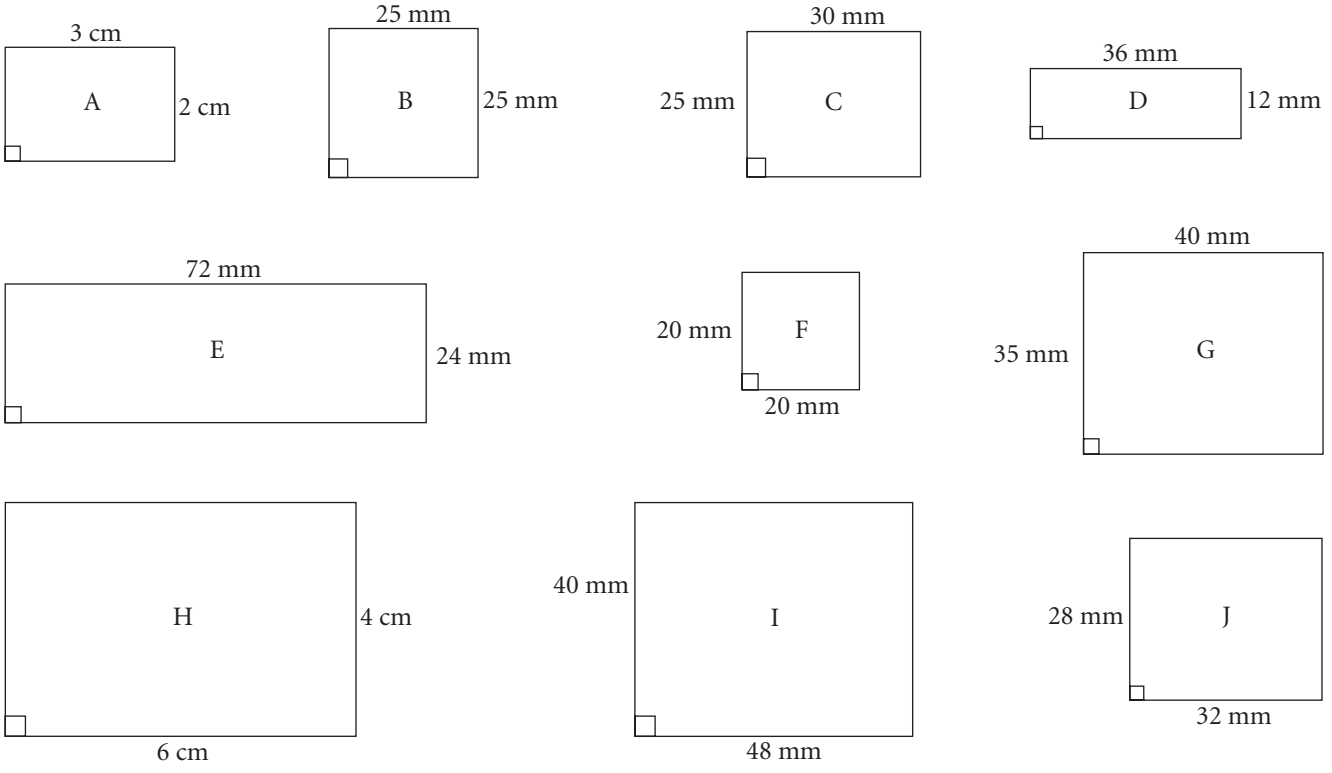
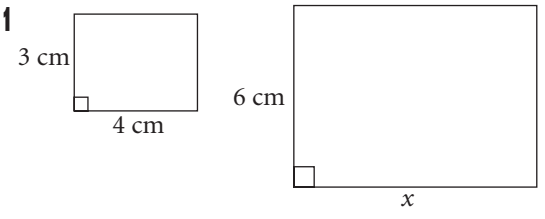
B Use the information on each diagram to match pairs of similar figures and state the ratio of corresponding sides.



C In each pair of similar figures, use the information on the diagrams to find the length of the side *x*.
(These diagrams are *not* drawn to scale.)

1



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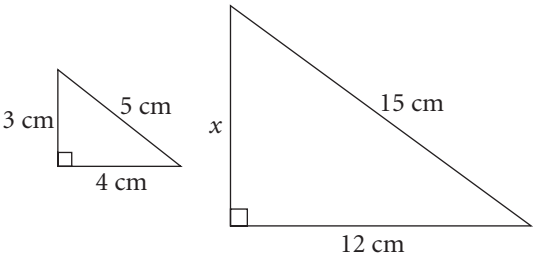

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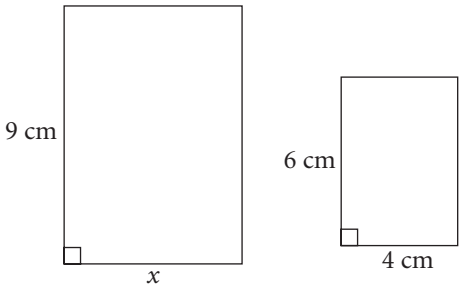

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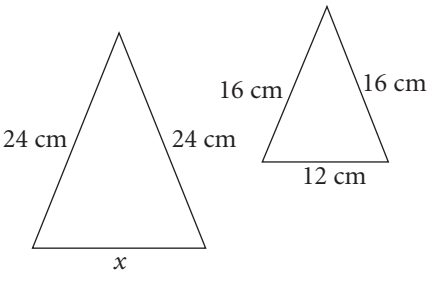

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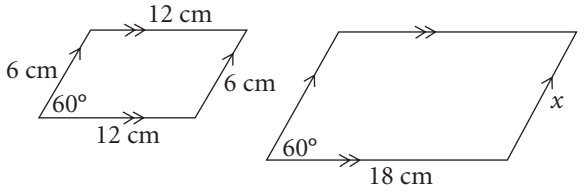

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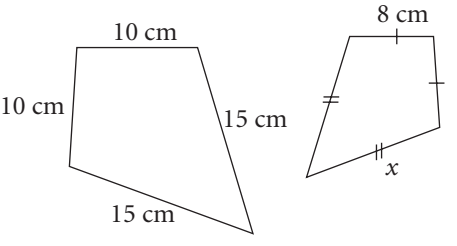

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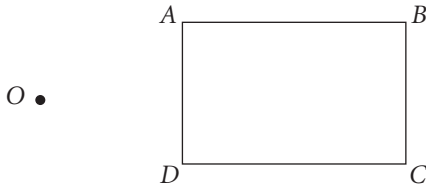
The enlargement transformation 6.1

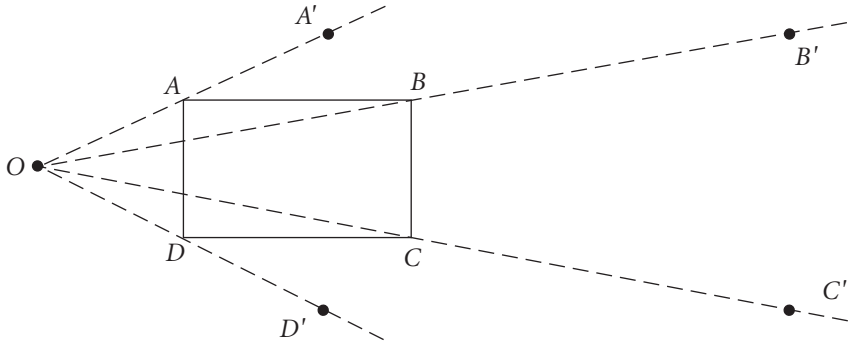
A We can construct similar figures using this method to enlarge or reduce the size of a given figure.

1 **a** Copy this rectangle on to your working paper. (It is often helpful to use a sheet of 5 mm grid paper.) Select a point O as the centre of enlargement. Label the vertices A, B, C and D.



Follow these steps to enlarge the rectangle by a factor of 2.

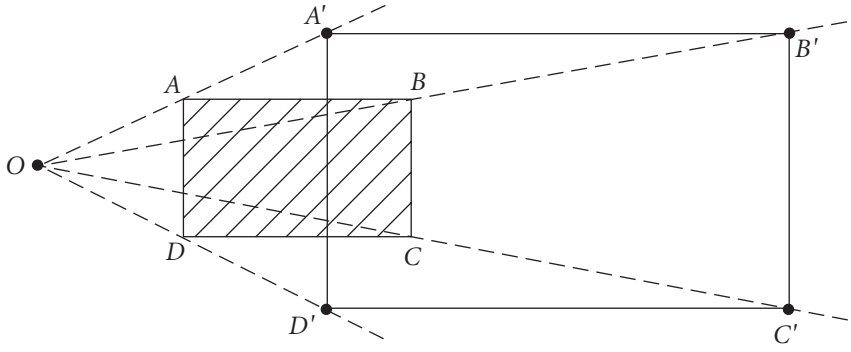
Step 1: Draw a construction line from O through each vertex of the figure.



Step 2: Measure the distance OA. Mark the point A' so that $OA' = 2 \times OA$.

Step 3: Measure the distance OB. Mark the point B' so that $OB' = 2 \times OB$.
Then mark the points C' and D' so that $OC' = 2 \times OC$ and $OD' = 2 \times OD$.

Step 4: Join the points A', B', C' and D' to form the new rectangle $A'B'C'D'$, the image of $ABCD$ in this transformation.



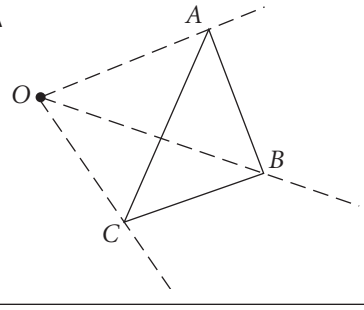
b Measure the lengths of the sides of each rectangle.

c Does $A'D' = 2 \times AD$ and $A'B' = 2 \times AB$?

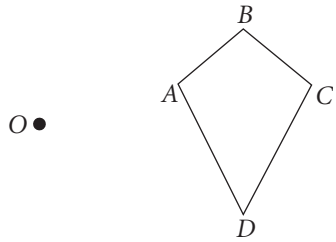
d Has each side of the rectangle $ABCD$ been multiplied by the enlargement factor 2?

2 Copy each figure and select a point O as the centre of enlargement.
Follow Steps 1 to 4 in question 1a to enlarge each figure by a factor of 2.

a



b



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B Enlarging a triangle by a factor of 3.

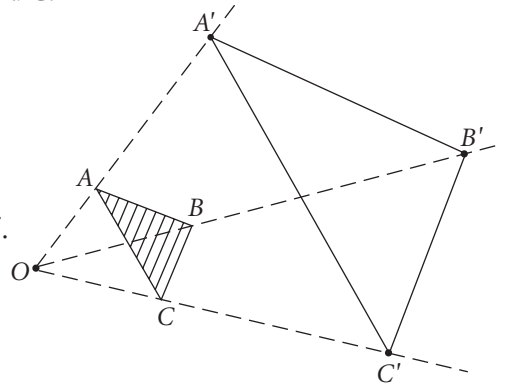
1 **a** Draw any triangle on your working paper and label the vertices A, B and C.

Step 1: Choose a point O as the centre of enlargement.

Step 2: Draw construction lines from O through each of the vertices A, B and C.

Step 3: Mark the positions of A', B' and C' so that $OA' = 3 \times OA$, $OB' = 3 \times OB$ and $OC' = 3 \times OC$.

Step 4: Join the points A', B' and C' to form the new triangle $A'B'C'$.

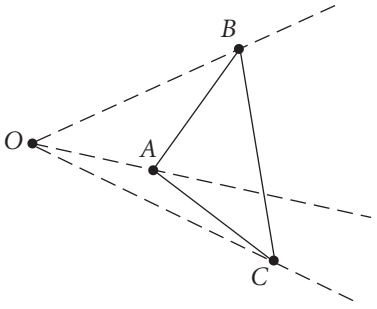


b Measure the lengths of the sides of each triangle.

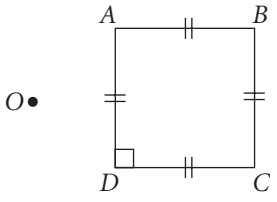
c Has each side of the triangle ABC been multiplied by the enlargement factor 3?

2 Copy each figure. Follow Steps 1 to 4 in question 1a to enlarge each figure by a factor of 3.

a



b

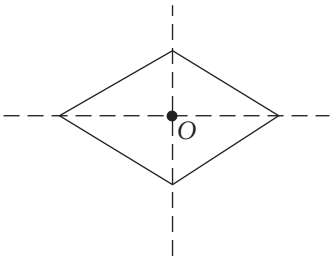


C

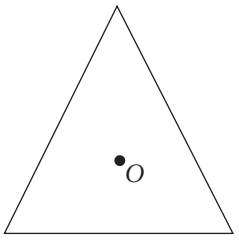
The centre of enlargement may be any point chosen outside, inside or on the figure.

1 Copy each figure and use a point O inside the figure as the centre of enlargement. Enlarge each figure by a factor of 2.

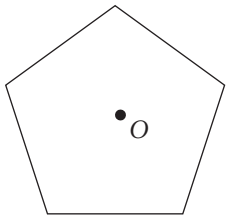
a



b

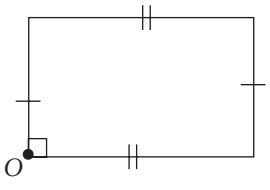


c

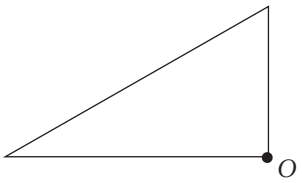


2 Copy each figure and enlarge it by a factor of 3, using the point O as the centre of enlargement. (The position of the point O will not change.)

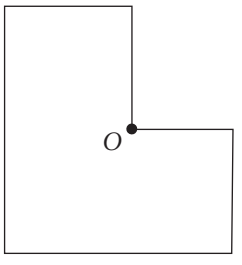
a



b



c



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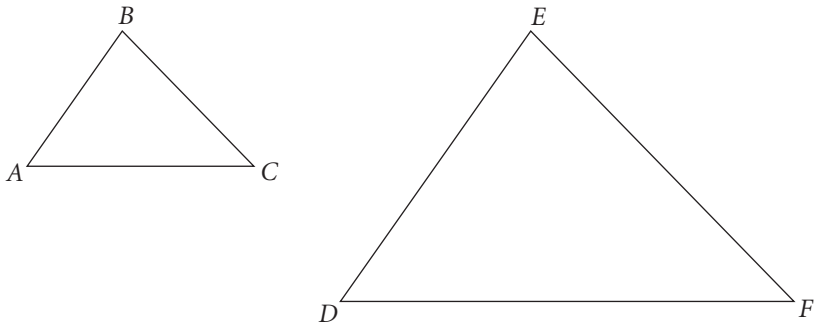
Similar triangles 7.1

A



ΔABC has been enlarged to give ΔDEF.
These two triangles are similar. We write $\Delta ABC \,|||\, \Delta DEF$ or $\Delta ABC \sim \Delta DEF$.

1 Angles in matching positions are the same size. Complete these statements: $\angle A =$ _____, $\angle B =$ _____, $\angle C =$ _____

2 Matching sides (sometimes called corresponding sides) are opposite equal-sized angles. Measure the length of each side of these triangles, to the nearest millimetre.

3 Calculate:

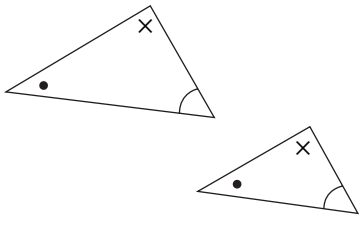
a $\frac{AB}{DE}$ **b** $\frac{BC}{EF}$ **c** $\frac{AC}{DF}$

4 What is the ratio of the corresponding sides of ΔABC to ΔDEF?

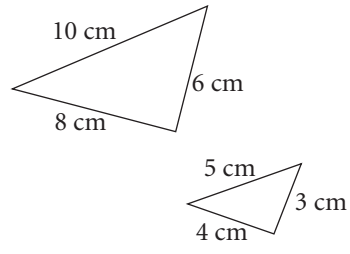
B

The following tests allow us to check if two triangles are similar:

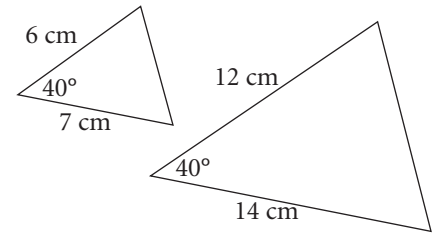
- Two triangles are similar if matching angles are equal. (We say they are equiangular.)



- Two triangles are similar if the corresponding sides are in the same ratio.

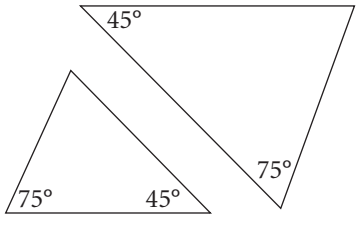


- Two triangles are similar if two pairs of corresponding sides are in the same ratio and the included angles are equal.

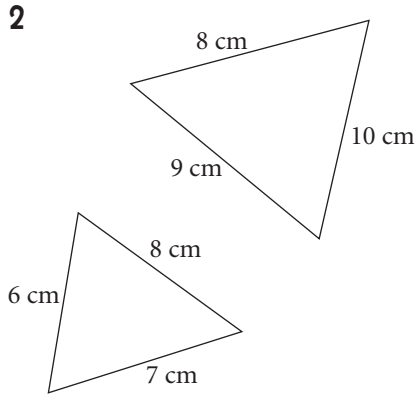


Are the triangles in each pair similar?

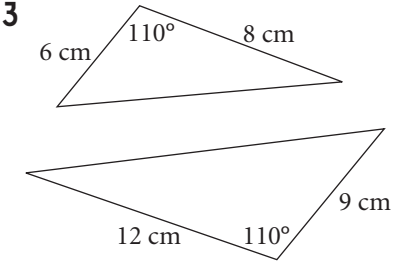
1



2



3



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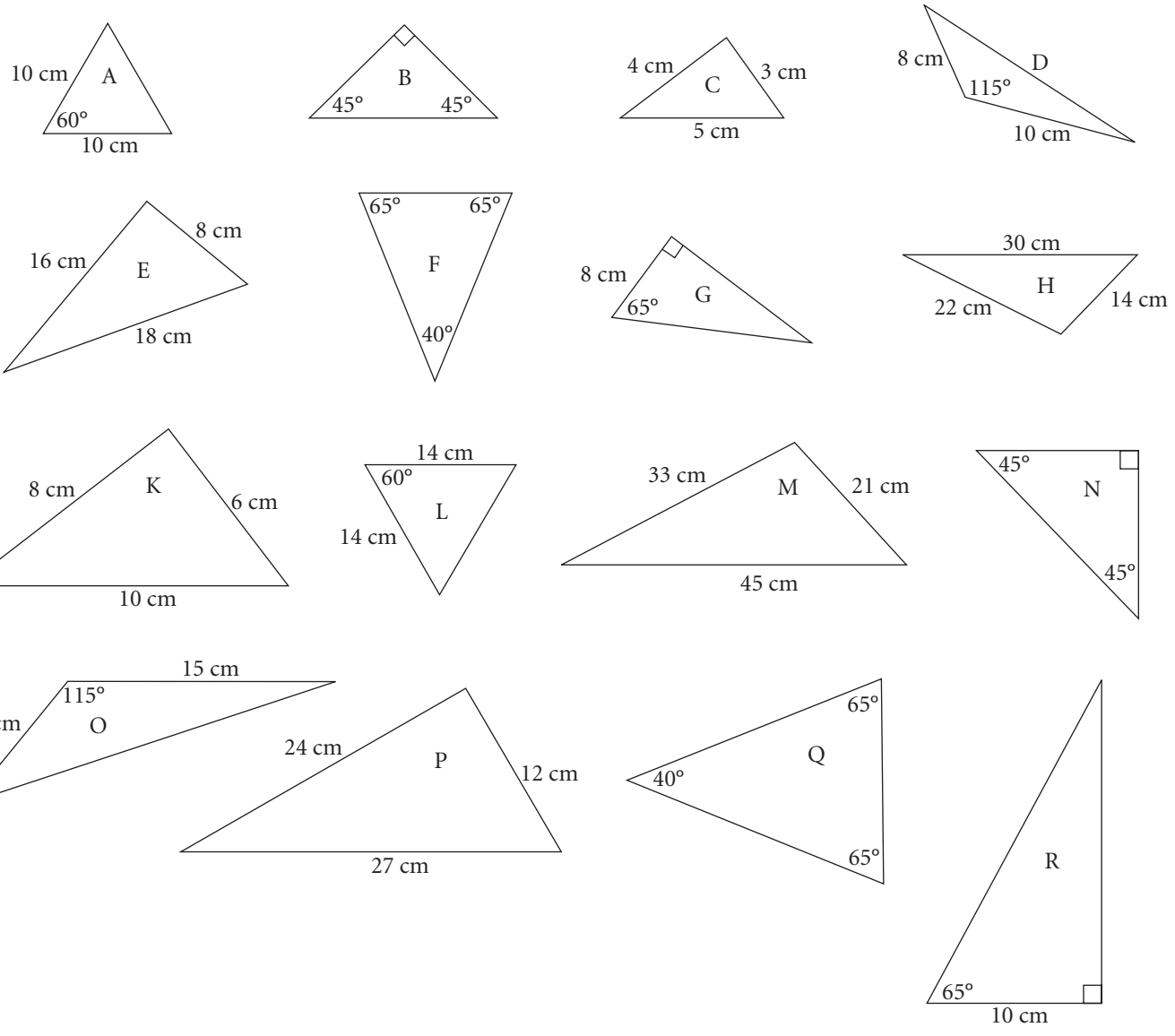
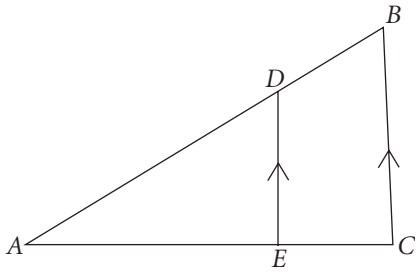
4 Each of the triangles labelled A to H is similar to one of the triangles labelled K to R. Match the pairs of similar triangles.



C

Complete these proofs that two triangles are similar.

1 In the triangle *ABC*, line *DE* is drawn parallel to side *BC*. Prove that Δ*ABC* and Δ*ADE* are similar.



In the Δs *ABC* and *ADE*,

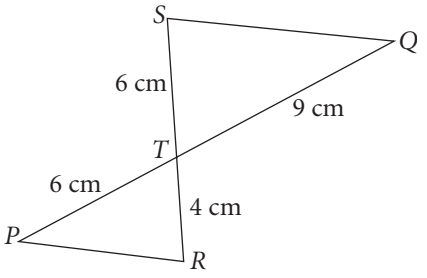
$\angle BAC$ is _____

$\angle ABC =$ _____ (equal corresponding angles, $BC \parallel DE$)

$\angle ACB =$ _____ (_____)

$\therefore$ Δ*ABC* is similar to Δ*ADE* (they are _____)

2 The straight lines *PQ* and *RS* intersect at the point *T*. Prove that Δ*PRT* and Δ*QST* are similar.



In the Δs *PRT* and *QST*,

$\frac{PT}{QT} = \frac{6}{\square} = \frac{2}{\square}$ and $\frac{RT}{ST} = \frac{\square}{\square} = \frac{\square}{\square}$

$\angle PTR =$ _____ (_____)

$\therefore$ Δ*PRT* is similar to Δ*QST* (_____)

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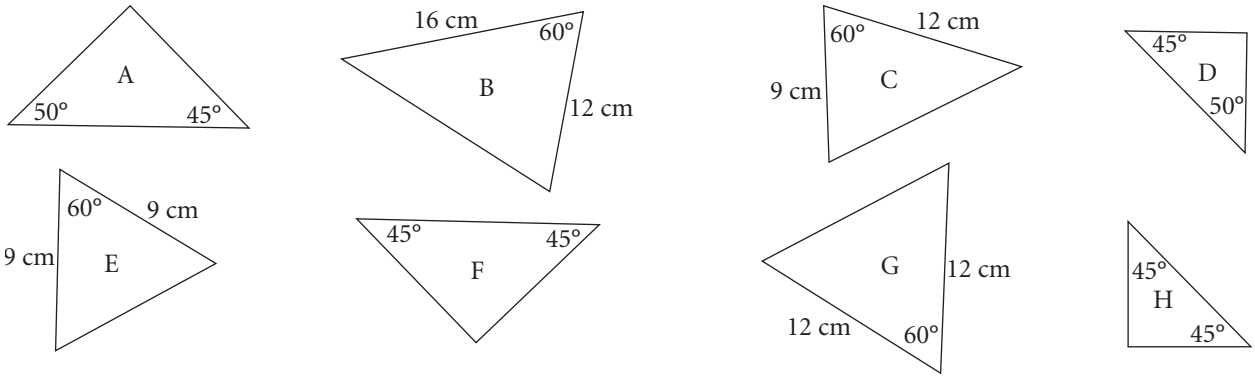
Using similar triangles 8.1

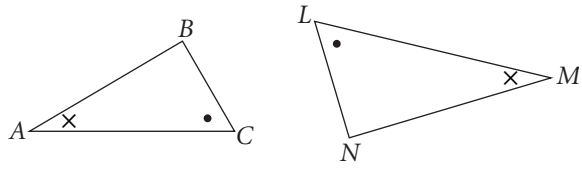
A

1 Match pairs of similar triangles.



2 The triangles *ABC* and *LMN* are similar.
Complete the ratios of corresponding sides.

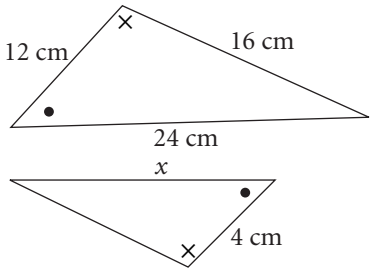
$$\frac{AB}{\square} = \frac{\square}{LN} = \frac{AC}{\square}$$



B

Example:

These two triangles are similar.
Find the value of x.



Solution:

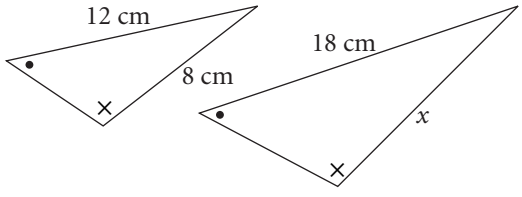
$$\frac{x}{24} = \frac{4}{12} \text{ (ratio of corresponding sides)}$$

$$\therefore x = 24 \times \frac{4}{12}$$

$$= 8 \text{ cm}$$

The triangles in each pair are similar. Calculate the value of each pronumeral.

1



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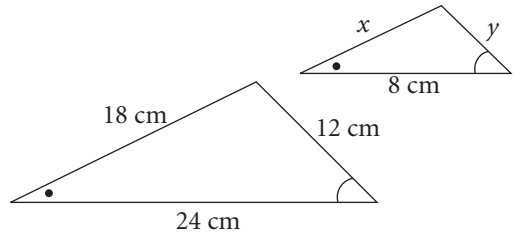

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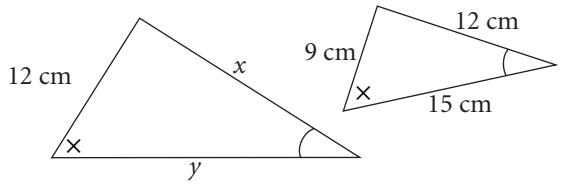
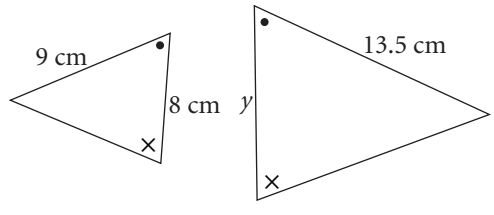


4



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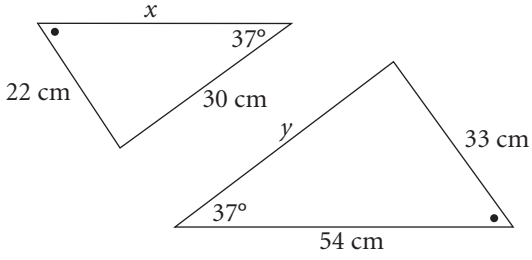

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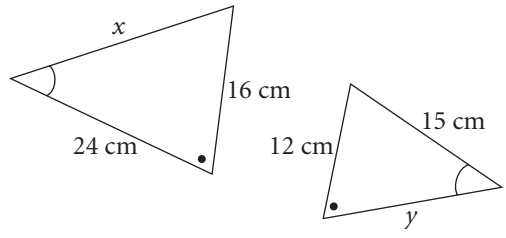

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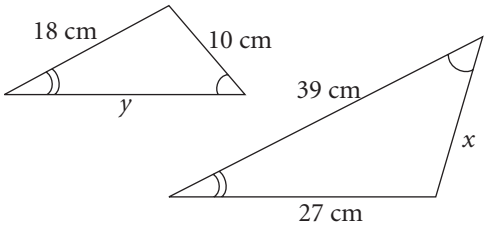
7



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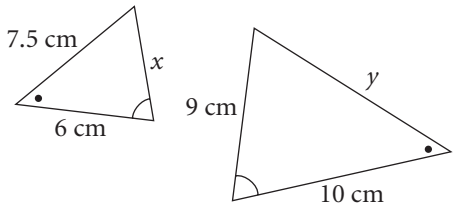

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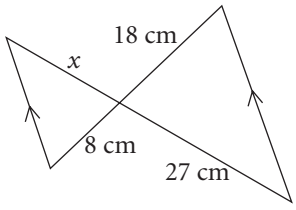

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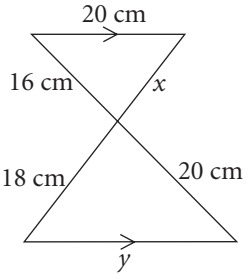
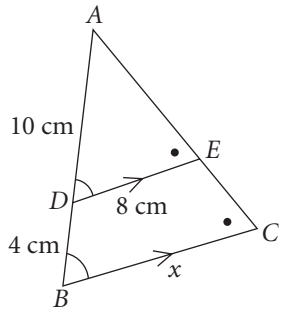


C

Example:

The triangles *ABC* and *ADE* are similar.
Find the value of *x*.



Solution:

$\frac{BC}{DE} = \frac{AB}{AD}$ (ratio of corresponding sides)

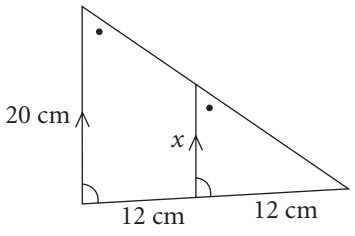
$\therefore \frac{x}{8} = \frac{14}{10}$

$x = 8 \times \frac{14}{10}$

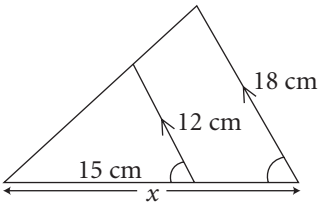
$= 11.2\,\text{cm}$

Find the values of the pronumerals in the following diagrams.

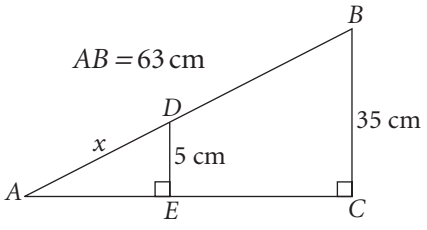
1



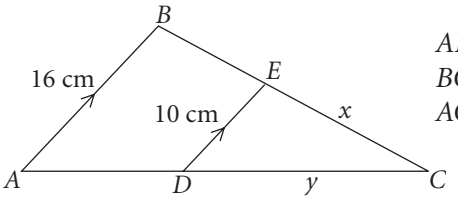
2



3

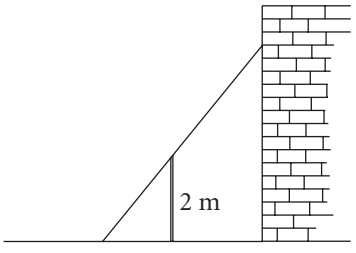


4

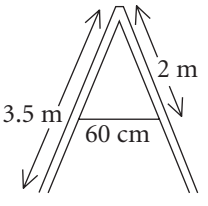


$AB \parallel DE$
$BC = 20\,\text{cm}$
$AC = 26\,\text{cm}$

5 A ladder leaning against a wall touches the top of a fence 2 metres high. The fence is 3 metres from the wall. If the foot of the ladder is 1.5 metres from the fence, calculate how far the ladder will reach up the wall.



6 A 3.5-metre step ladder has a brace 60 cm long placed 2 metres from the top. How far apart are the feet of the ladder?



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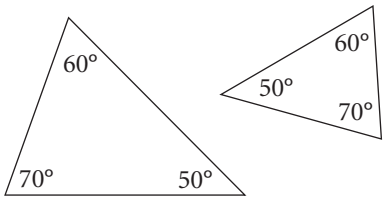
Similar triangles and reasoning 9.1

A

Are the following pairs of triangles similar?

1



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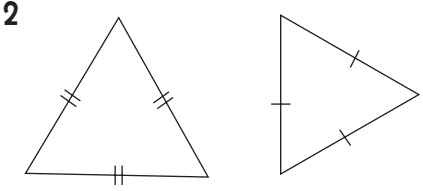

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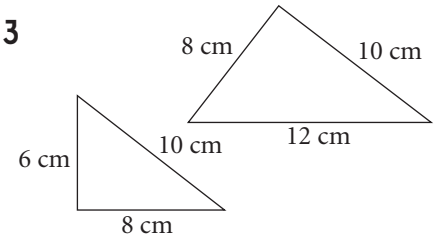
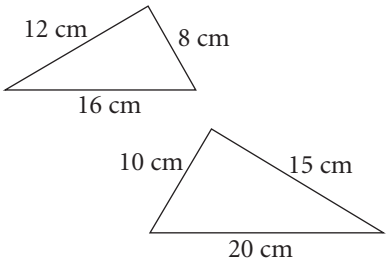
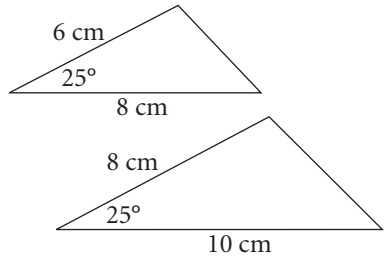


4



5



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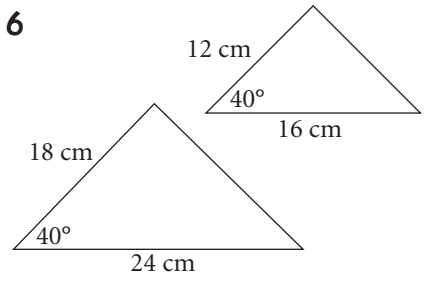
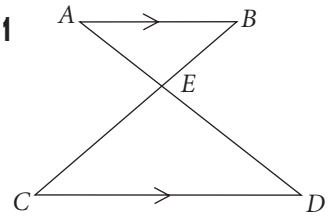


B

Name two similar triangles in each of the following diagrams. Briefly explain why they are similar.

1



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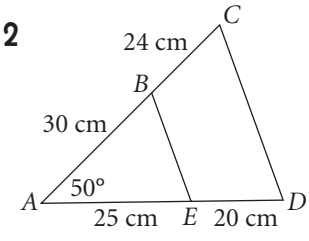

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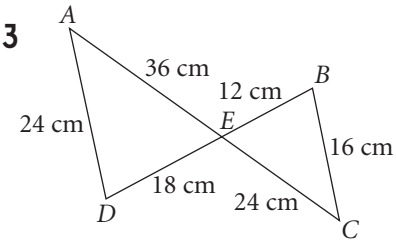
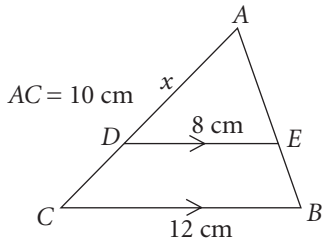


Example:

In the triangle ABC, $DE \| CB$ and $AC = 10$ cm. Prove that ΔACB is similar to ΔADE and find the value of x.



Solution:

In the Δs ACB and ADE, $\angle BAC$ is common, $\angle ACB = \angle ADE$ (equal corresponding angles, $CB \| DE$) and $\angle ABC = \angle AED$.

$\therefore$ ΔACB is similar to ΔADE (equiangular)

$\therefore$ $\frac{AD}{AC} = \frac{DE}{CB}$ (ratio of corresponding sides)

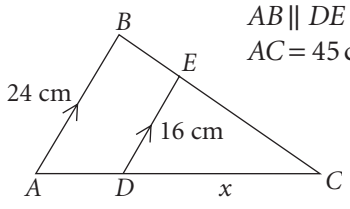
$\therefore$ $\frac{x}{10} = \frac{8}{12}$

$\therefore$ $x = 10 \times \frac{8}{12}$

$= 6.7$ cm (to 1 decimal place)

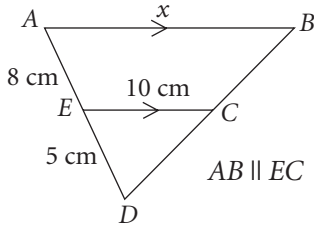
4 **a** Prove that ΔABC is similar to ΔDEC.

b Find the length of DC.



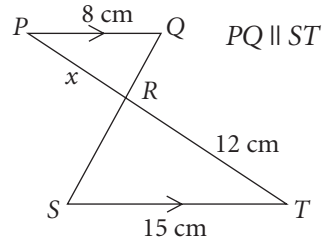
5 **a** Prove that ΔABD is similar to ΔECD.

b Find the length of AB.



6 **a** Prove that ΔPQR is similar to ΔTSR.

b Find the length of PR.



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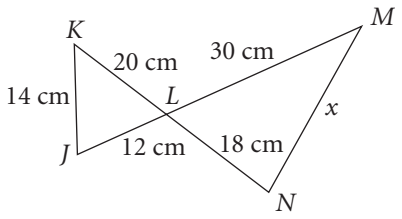
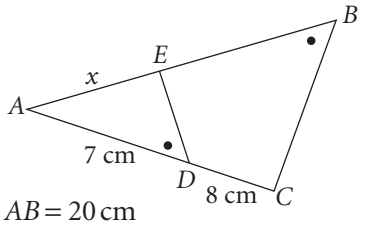
7 **a** Prove that ΔJKL is similar to ΔNML.

b Find the length of MN.



8 **a** Prove that ΔABC is similar to ΔADE.

b Find the length of AE.

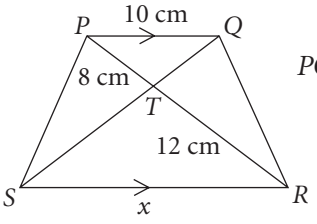


$AB = 20$ cm

$\angle ABC = \angle ADE$

9 **a** Prove that ΔPQT is similar to ΔRST.

b Find the length of SR.



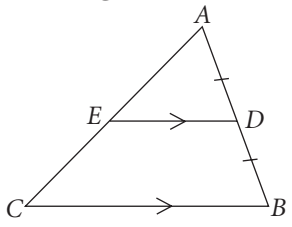
$PQ \parallel SR$

C

1 D is the midpoint of AB and $ED \parallel CB$. Prove that:

a ΔABC is similar to ΔADE

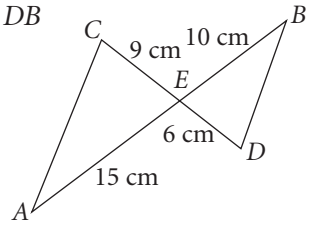
b $AE = EC$



2 Straight lines AB and CD intersect at the point E. Prove that:

a ΔACE is similar to ΔBDE

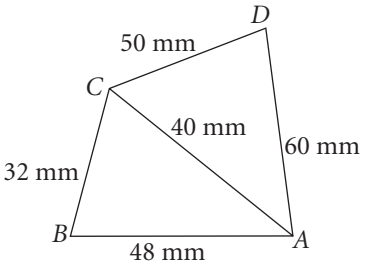
b $AC \parallel DB$



3 Prove that:

a ΔABC is similar to ΔDAC

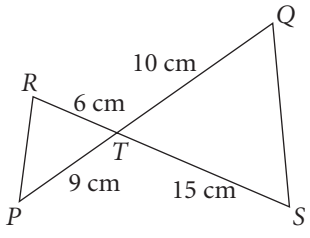
b $\angle BCA = \angle DCA$



4 Straight lines PQ and RS intersect at the point T.

a Prove that ΔPRT is similar to ΔSQT.

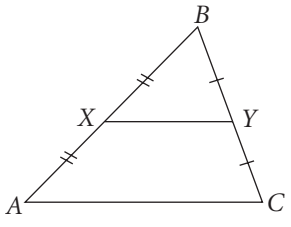
b Is $PR \parallel QS$?



5 X and Y are the midpoints of AB and BC respectively. Prove that:

a ΔABC is similar to ΔXBY

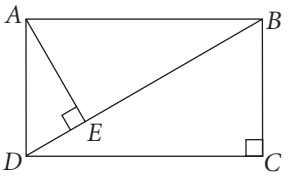
b $AC \parallel XY$



6 $ABCD$ is a rectangle. AE is drawn perpendicular to the diagonal DB. Prove that:

a ΔADE is similar to ΔDBC

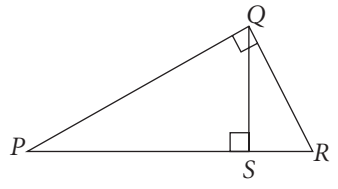
b $AD \times DC = DB \times AE$



7 PQR and PSQ are right-angled triangles. Prove that:

a ΔPQR is similar to ΔPSQ

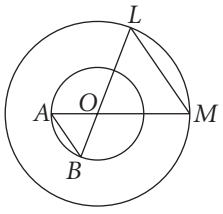
b $PQ^2 = PS \times PR$



8 O is the centre of the two circles where the radius of the larger circle is twice the radius of the smaller circle.

a Prove that ΔAOB is similar to ΔLOM.

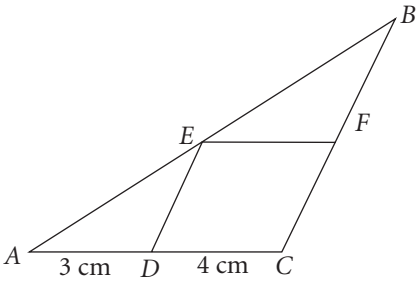
b What is the ratio $AB : LM$?



9 $CDEF$ is a parallelogram. $AD = 3$ cm, $DC = 4$ cm.

a Prove that ΔADE is similar to ΔEFB.

b What is the ratio $CF : FB$?



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Areas of similar figures 10.1

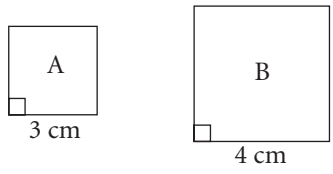
A Investigation

1 Compare these two squares. Square B is an enlargement of square A. They are similar figures.

a What is the ratio of their sides?

b Calculate the area of each square.

c What is the ratio of their areas?



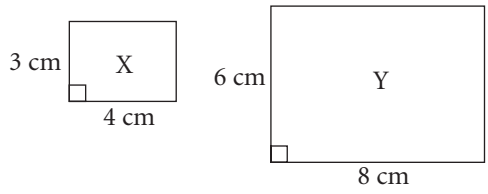
2 Two rectangles X and Y are similar.

a What is the ratio of their lengths?

b What is the ratio of their breadths?

c Calculate the area of each rectangle.

d What is the ratio of their areas?



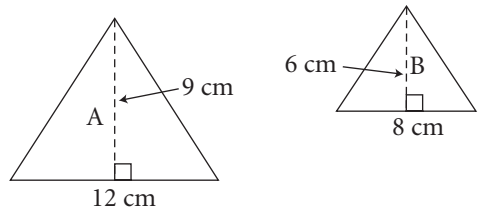
3 These two triangles are similar.

a What is the ratio of their heights?

b What is the ratio of the lengths of their bases?

c Calculate the area of each triangle.

d What is the ratio of their areas?



From these investigations, we can deduce the rule: *In any pair of similar figures, if corresponding lengths are in the ratio a : b then their areas are in the ratio $a^2 : b^2$.*

Example:

Two similar figures have sides in the ratio 2 : 3. If the area of the larger figure is 72 cm^2, what is the area of the smaller figure?

Solution: Ratio of sides = 2 : 3

$\therefore$ Ratio of areas = $2^2 : 3^2$

= 4 : 9

$\therefore$ Area : 72 = 4 : 9

OR $\frac{\text{Area}}{72} = \frac{4}{9}$

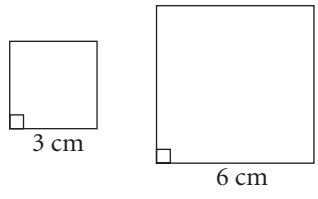
$\therefore$ Area = $72 \times \frac{4}{9} = 32\,\text{cm}^2$

B

1 Two squares have sides measuring 3 cm and 6 cm.

a What is the ratio of their sides?

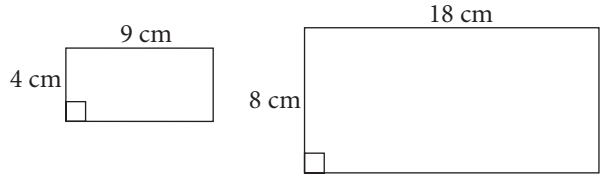
b What is the ratio of their areas?



2 **a** Explain why these two rectangles are similar.

b What is the ratio of corresponding sides?

c What is the ratio of their areas?



3 Two similar rectangles have corresponding sides in the ratio 3 : 5. What is the ratio of their areas?

4 Two squares have sides measuring 6 cm and 8 cm. What is the ratio of:

a the lengths of their sides?

b their areas?

5 The corresponding sides of two similar triangles are in the ratio 2 : 3. If the larger triangle has an area of 144 cm^2, what is the area of the smaller triangle?

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6 The diameters of two circles are in the ratio 3 : 4. The area of the larger circle is 44.8 cm^2. What is the area of the smaller circle?

7 The corresponding sides of two similar pentagons are in the ratio 3 : 5. The smaller pentagon has an area of 36 cm^2. What is the area of the larger pentagon?

8 The corresponding sides of two similar quadrilaterals are in the ratio 3 : 4. What is the area of the larger quadrilateral if the area of the smaller quadrilateral is 22.5 cm^2?

9 If you double the lengths of each side in a regular hexagon, by what factor does this increase the area?

10 Two similar rectangles have areas 80 cm^2 and 45 cm^2.

a What is the ratio of their areas, in its simplest form?

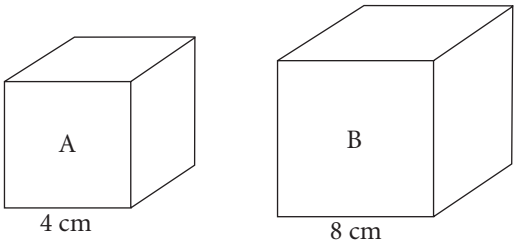
b What is the ratio of the lengths of corresponding sides?

11 The areas of two similar triangles are in the ratio 25 : 9. What is the ratio of corresponding lengths?

12 Two similar triangles have areas in the ratio 4 : 9. If one side of the smaller triangle is 15 cm long, what is the length of the corresponding side in the larger triangle?

C

1 Two cubes A and B have edges measuring 4 cm and 8 cm respectively. Cube B is an enlargement of cube A. They are *similar solids*.



a What is the ratio of corresponding lengths?

b What is the area of each face of cube A? What is the total surface area of this cube?

c What is the area of each face of cube B? What is the total surface area of this cube?

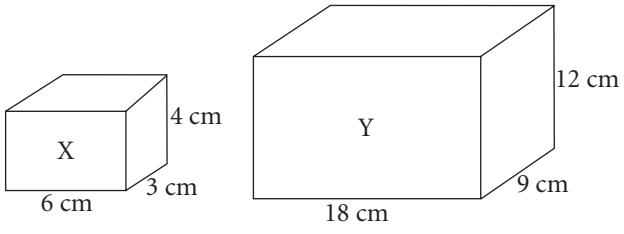
d What is the ratio of the area of one face of cube A to the area of one face of cube B?

e What is the ratio of their surface areas? (Simplify where possible.)

f Calculate the volume of each cube.

g What is the ratio of their volumes?

2 The diagrams show two rectangular prisms. Prism Y is an enlargement of prism X. They are *similar solids*.



a What is the ratio of corresponding edges?

b Calculate the total surface area of the smaller prism.

c Calculate the total surface area of the larger prism.

d What is the ratio of their surface areas, in simplest form?

e Calculate the volume of the smaller prism.

f Calculate the volume of the larger prism.

g What is the ratio of their volumes, in simplest form?

3 Compare your results in questions 1 and 2 above.

a How is the ratio of the *surface areas* of similar solids related to the lengths of corresponding edges?

b How is the ratio of the *volumes* of similar solids related to the lengths of corresponding edges?

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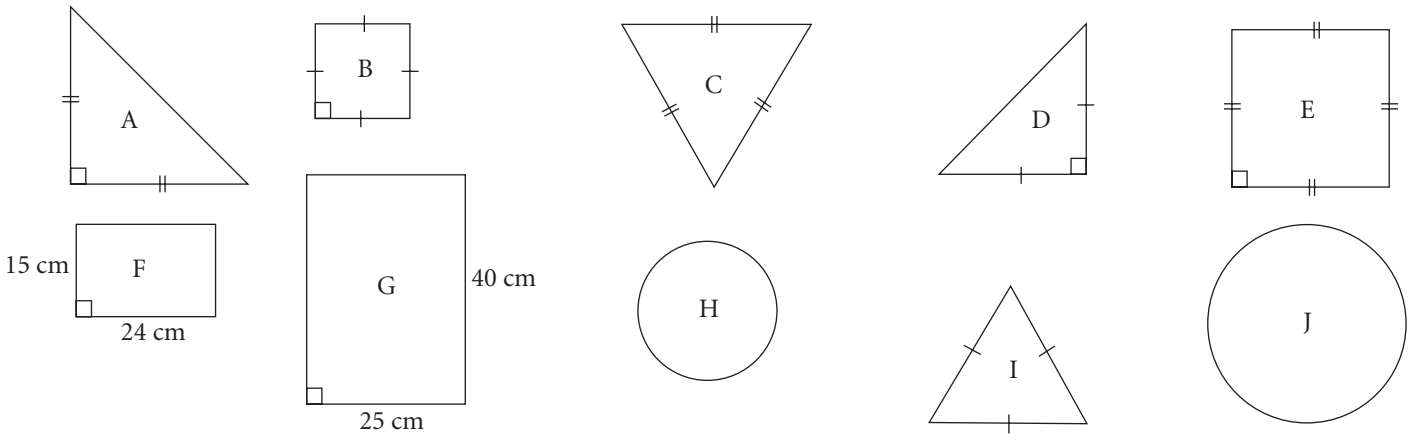
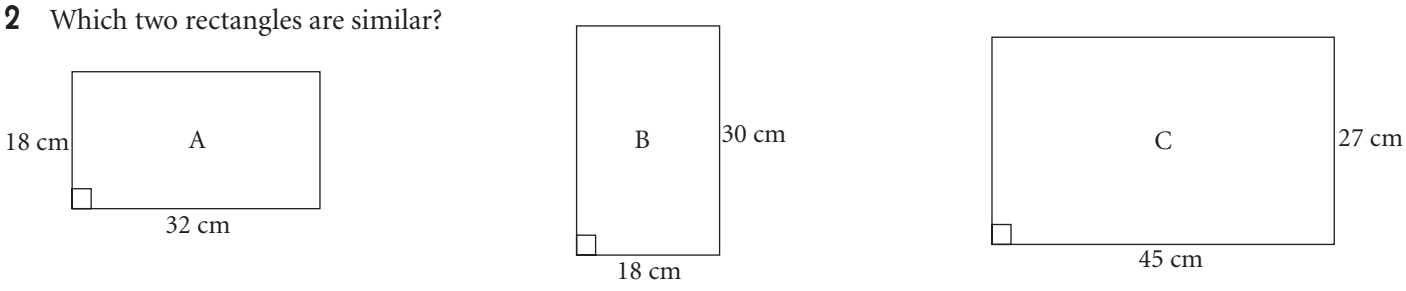
Review — Similar figures 11.1

A

1 Match pairs of similar figures.

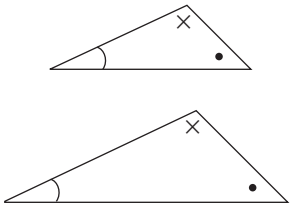


2 Which two rectangles are similar?

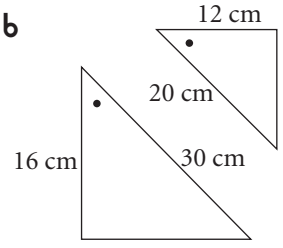


3 Are the following pairs of triangles similar?

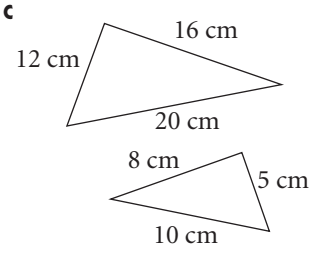
a



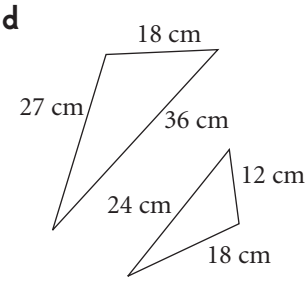
b



c



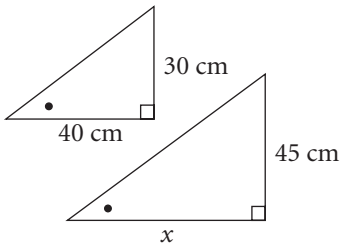
d



B

Each pair of triangles is similar. Find the value of each pronumeral.

1



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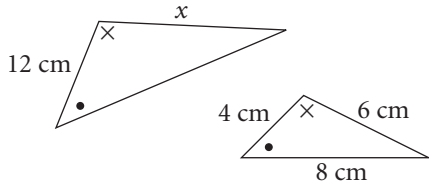

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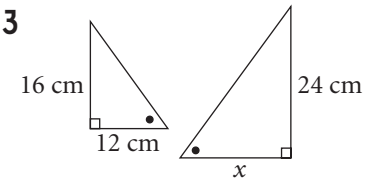

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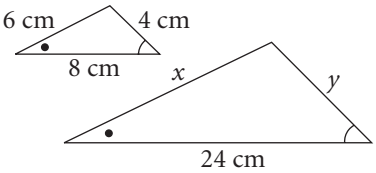

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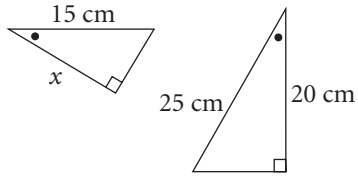

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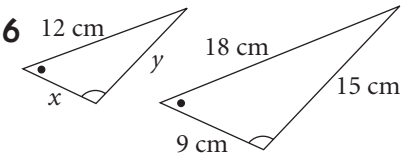

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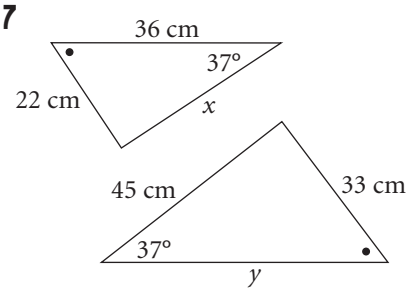
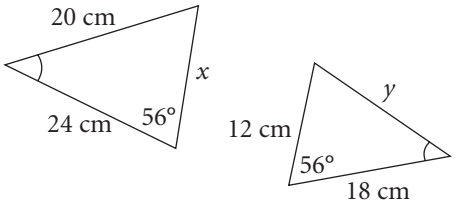
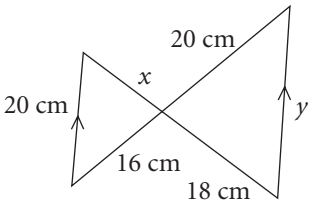


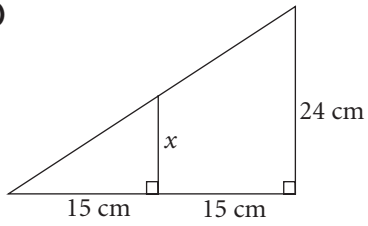
8



9



10

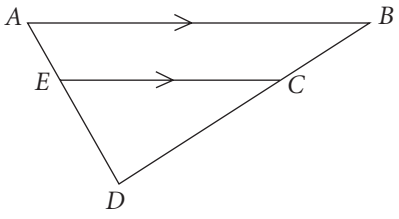


11 Two similar rectangles have corresponding sides in the ratio 3 : 4. What is ratio of their areas?

12 The corresponding sides of two similar triangles are in the ratio 2 : 3. If the smaller triangle has an area of 144 cm^2, what is the area of the larger triangle?

C

1 In this diagram, *EC* has been drawn parallel to *AB*. Complete this proof that Δ*ABD* is similar to Δ*ECD*.

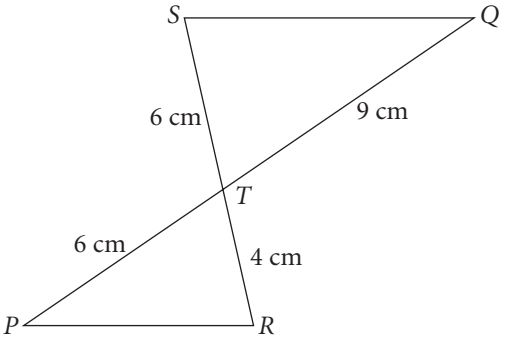


In the Δs *ABD* and *ECD*,
∠_____ is common.
∠*DAB* = _____ (_____)
∠*ABD* = _____ (_____)
∴ Δ*ABD* is similar to Δ*ECD* (_____)

2 The straight lines *PQ* and *RS* intersect at the point *T*.

a Prove that Δ*PRT* is similar to Δ*QST*.

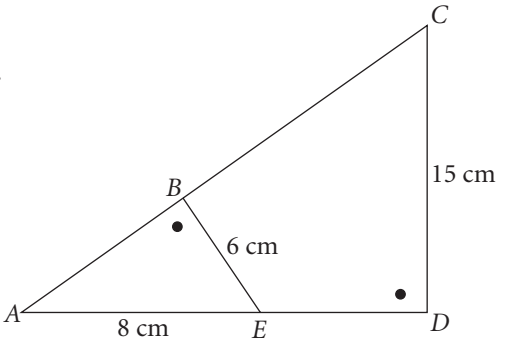
b Explain why *PR* is parallel to *SQ*.



3 In this diagram, ∠*ABE* = ∠*ADC*.

a Prove that Δ*ABE* is similar to Δ*ADC*.

b Find the length of *AC*.



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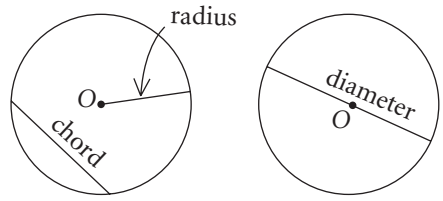
Circles 12.1

A

A circle is formed by the set of points in a plane that are all the same distance (the radius) from a fixed point, called the centre. The centre of a circle is usually marked as the point *O*.
A *chord* is an interval (or line segment) joining two points on the circle.
A *diameter* is a chord which passes through the centre of the circle. All the diameters of a given circle are the same length and twice the length of the radius.



1 Use a pair of compasses to construct a circle:

a with radius 4 cm
b with radius 45 mm
c with radius 55 mm
d with diameter 10 cm
e with diameter 70 mm
f with diameter 120 mm

2 Use a pair of compasses and the same point as centre to draw 3 circles with radius:

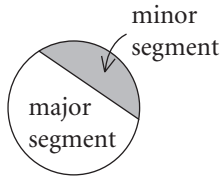
a 5 cm
b 6 cm
c 8 cm

These are called *concentric circles* because they have the same centre.

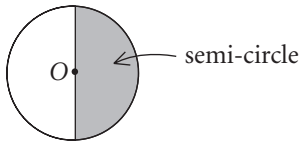
B

Parts of a circle

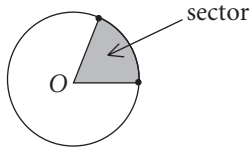
- A chord divides the region inside a circle into two *segments*. The larger region is called the *major segment* and the smaller region is called the *minor segment*.



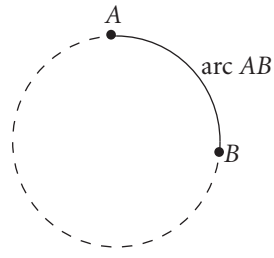
- A diameter divides a circle in half.
- Half a circle is called a *semicircle*.



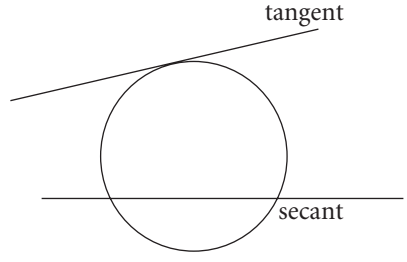
- A *sector* is the region inside a circle bounded by two radii and part of the circle.



- The *circumference* is the length of the boundary, or the distance around a circle.
- An *arc* is part of the boundary of a circle. An arc is named using its end points.

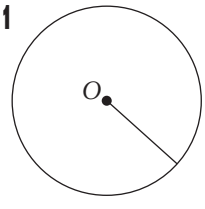


- A *tangent* is a straight line that touches a circle at one point only.
- A *secant* is a straight line that cuts through a circle at two points.



Name the feature or part of a circle indicated in each diagram.

1



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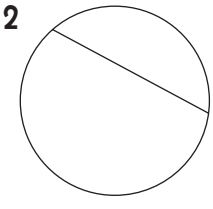

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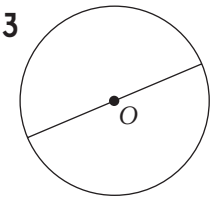

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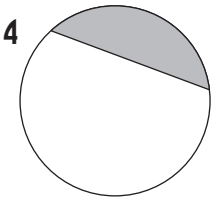

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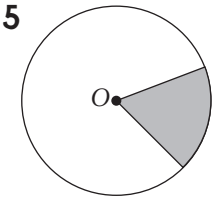

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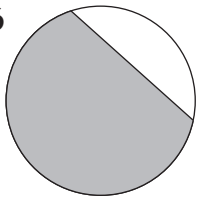

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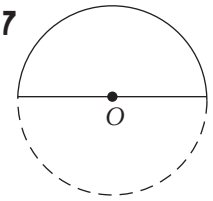

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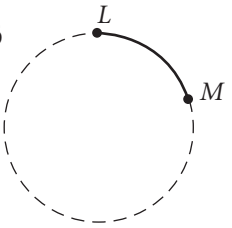

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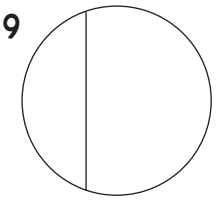

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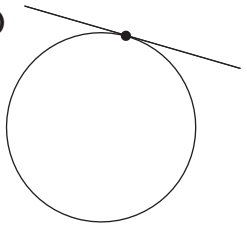

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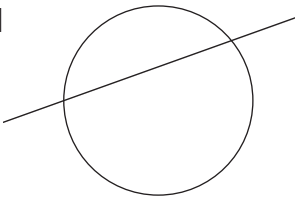

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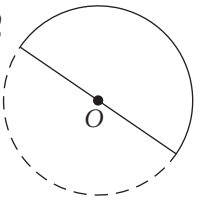

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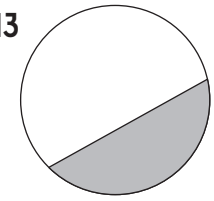

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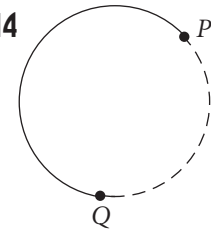

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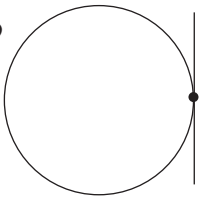

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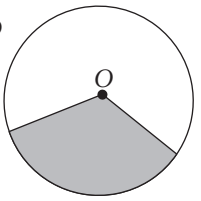
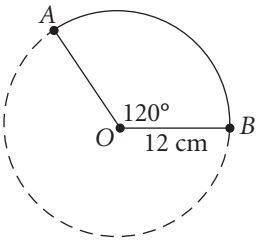


C

To calculate the *length of an arc*, use the size of the angle (θ) it subtends at the centre of the circle.

$$\text{length of arc} = \frac{\theta}{360} \times 2 \times \pi \times (\text{radius})$$

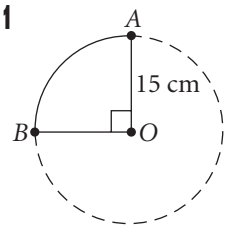


Example: arc $AB = \frac{120}{360} \times 2 \times \pi \times 12$

$= 25.1$ cm (to 1 decimal place)

Calculate the length of the arc *AB*, in each diagram, correct to one decimal place.

1



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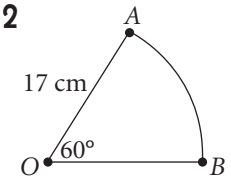

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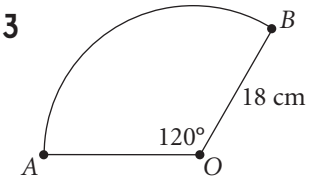

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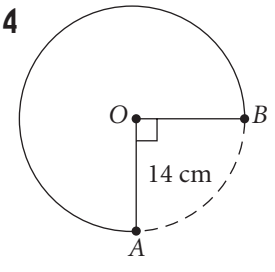

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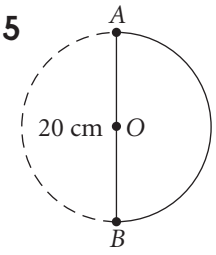

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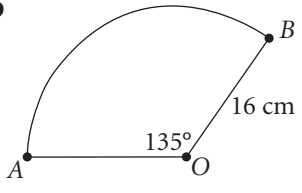

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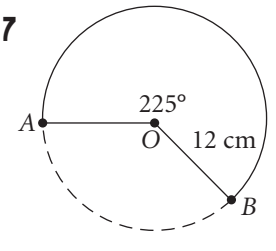

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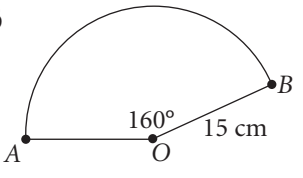

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Chord properties of circles 13.1

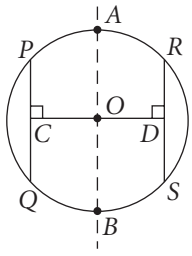
A

- Any *diameter* will be an *axis of symmetry* of a circle.
- On opposite sides of an axis of symmetry, matching intervals are the same length and matching angles are the same size.

Investigation

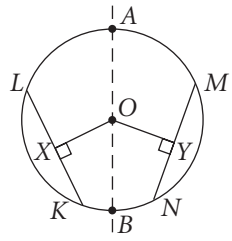
In each diagram, *AB* is an axis of symmetry and the point *O* is the centre of the circle. Complete the pairs of equal intervals and equal angles, which lie on opposite sides of *AB*.

1



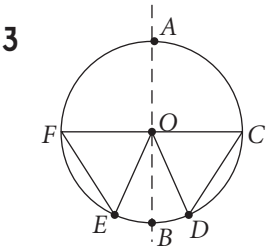
$PQ =$
$OC =$

2



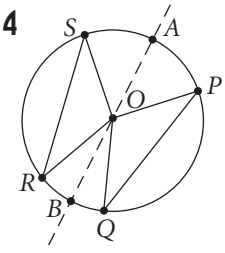
$KL =$
$OX =$

3



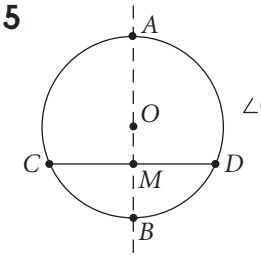
$CD =$
$\angle DOC =$
$OC =$
$OD =$

4



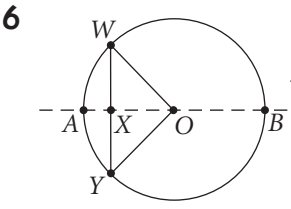
$PQ =$
$\angle POQ =$
$OP =$
$OQ =$

5



$CM =$
$\angle OMC =$

6



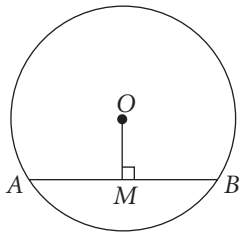
$OW =$
$XW =$
$\angle XWO =$

Notice that:

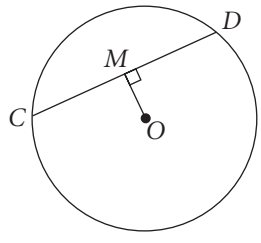
- equal chords are the same distance from the centre of the circle (refer to Q1 and Q2)
- equal chords subtend equal angles at the centre of the circle (refer to Q3 and Q4)
- equal angles, at the centre of a circle, stand on equal chords (refer to Q3 and Q4)
- a line drawn from the centre of a circle to the midpoint of a chord is perpendicular to the chord
- the perpendicular bisector of a chord passes through the centre of the circle (refer to Q5).

B In each diagram, the point *O* is the centre of the circle. (Diagrams are *not* drawn to scale.)

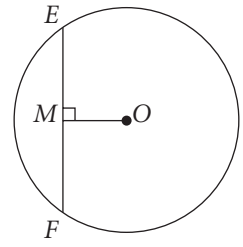
1 $AB = 16$ cm
Find the length of *MB*.



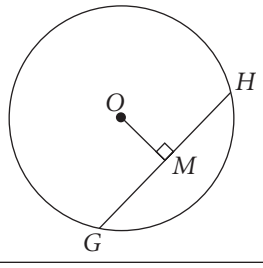
2 $CD = 28$ cm
Find the length of *CM*.



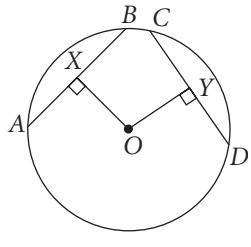
3 $EM = 12$ cm
Find the length of *EF*.



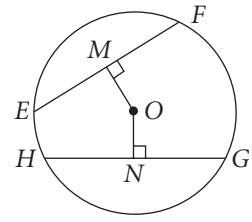
4 $MH = 9$ cm
Find the length of *GH*.



5 $AB = CD$, $OX = 6$ cm
Find the length of *OY*.



6 $EF = GH$, $OM = 8$ cm
Find the length of *ON*.



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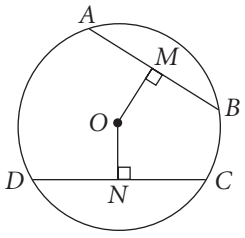
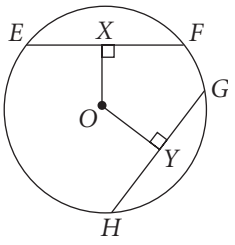
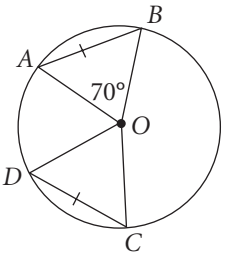
7 $ON = OM$, $AB = 13$ cm
Find the length of CD.



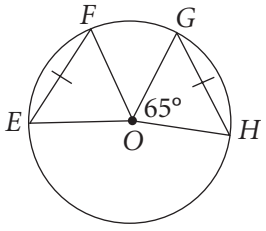
8 $OX = OY$, $EF = 20$ cm
Find the length of GH.



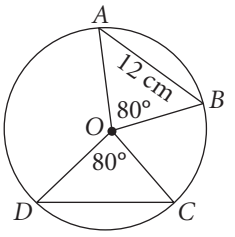
9 $AB = CD$, $\angle AOB = 70°$
Find the size of $\angle DOC$.



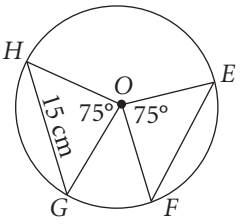
10 $EF = GH$, $\angle GOH = 65°$
Find the size of $\angle EOF$.



11 $\angle AOB = \angle DOC$, $AB = 12$ cm
Find the length of CD.

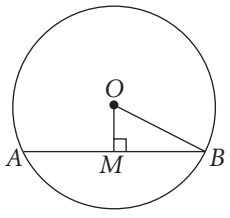


12 $\angle EOF = \angle GOH$, $GH = 15$ cm
Find the length of EF.

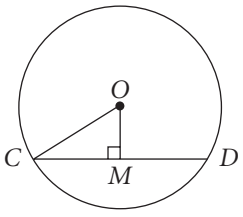


In the following questions, you will need to use Pythagoras' theorem.

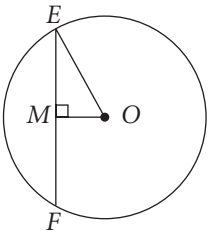
13 $OB = 26$ cm, $OM = 10$ cm
Find the length of MB.



14 $OC = 17$ cm, $OM = 8$ cm
Find the length of CD.

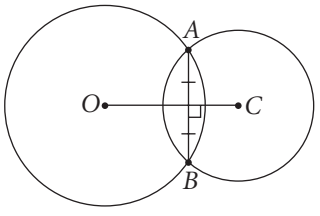


15 $ME = 8$ cm, $OM = 6$ cm
Find the length of OE.



C

When two circles intersect, the line joining their centres is the perpendicular bisector of the common chord. (It meets the common chord at right angles and cuts it in half.)



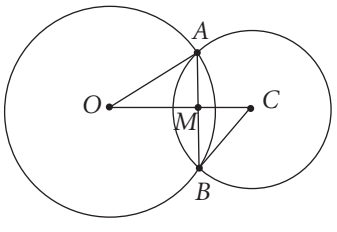
The points O and C are the centres of the two circles.
AB is the common chord.

1 O and C are the centres of the two circles which meet at the points A and B. If $OA = 13$ cm, $CB = 10$ cm and $AB = 12$ cm, calculate the length of:

a CM

b OM (to 1 decimal place)

c OC, the distance between the two centres.

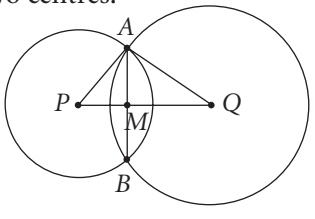


2 P and Q are the centres of two circles which meet at the points A and B. The length of the common chord AB is 24 cm. $AP = 13$ cm and $AQ = 15$ cm. Calculate the length of:

a PM

b MQ

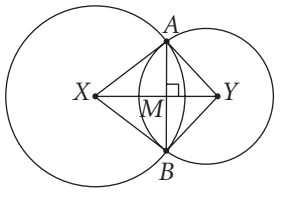
c PQ, the distance between the two centres.



3 X and Y are the centres of two circles which meet at the points A and B. AB and XY meet at the point M. Given $XM = 36$ cm, $MY = 20$ cm and $AX = 39$ cm, calculate:

a the length of the common chord AB

b the area of the kite $YBXA$.



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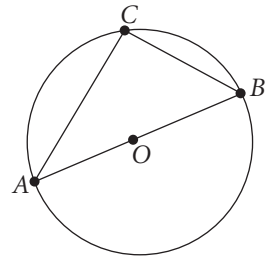
Angle properties of circles 14.1

A

Investigation You will need a sheet of paper to construct your own diagrams. (Draw a new diagram for each question.)



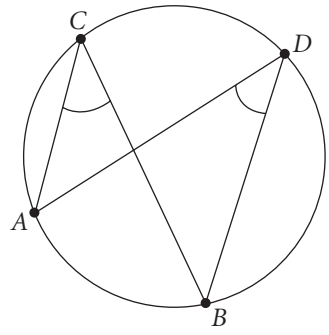
1 **a** Draw a circle, marking the centre point *O*.

b Draw a diameter *AB* on your circle.

c Select any point on the circumference and label it point *C*.

d Draw the lines *AC* and *BC* and measure the size of $\angle ACB$.

e Repeat the last two steps with a different point *C* on the circumference of your circle.

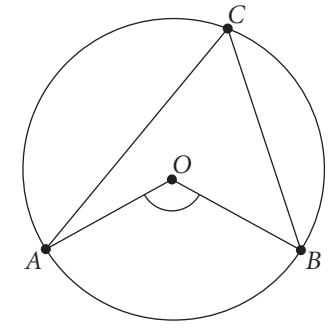


2 **a** Draw a circle and select any two points on the circumference. Label these points *A* and *B*.

b Select a third point *C* on the circumference and draw the lines *AC* and *BC*. Measure the size of $\angle ACB$.

c Select a fourth point *D* on the circumference (in the same segment as point *C*) and draw the lines *AD* and *BD*. Measure the size of $\angle ADB$.

d Does $\angle ACB = \angle ADB$?



3 **a** Draw a circle, marking the centre point *O*.

b Select any two points on the circumference and label these points *A* and *B*.

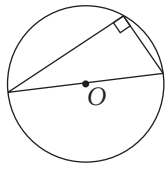
c Draw the lines *OA* and *OB* and measure the size of $\angle AOB$.

d Select a point *C* on the circumference of the circle. Draw the lines *AC* and *BC* and measure the size of $\angle ACB$.

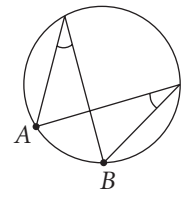
e Compare the size of $\angle ACB$ and $\angle AOB$. (Be careful to measure the angles *ACB* and *AOB* which stand on the same arc.)

You have demonstrated these properties:

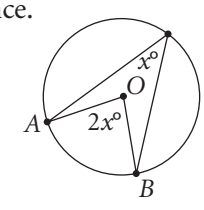
- The angle in a semicircle is a right angle. (An angle at the circumference, standing on a diameter, is a right angle.)



- Angles at the circumference, standing on the same arc, are equal in size.



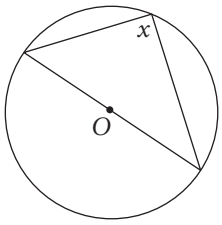
- The angle subtended by an arc at the centre of a circle is twice the size of an angle subtended by the same arc at the circumference.



B

Find the value of the pronumeral *x* in each diagram. (*O* is the centre of each circle.)

1



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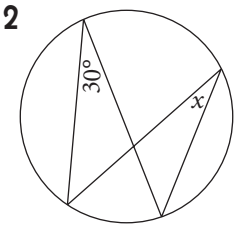

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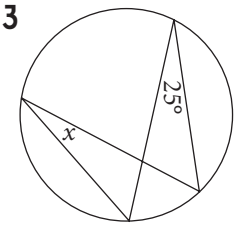

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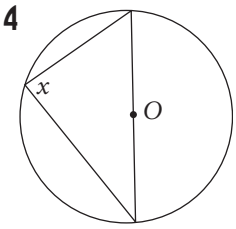

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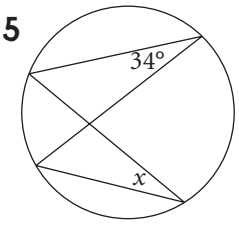
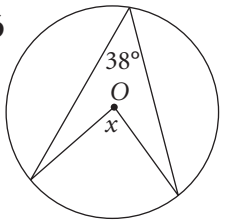
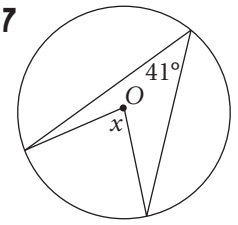
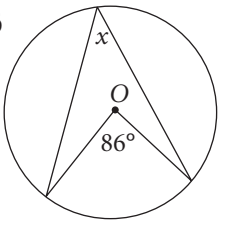
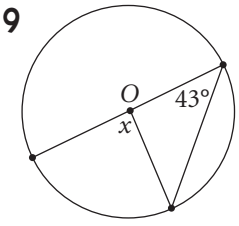
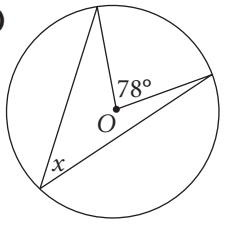
5 

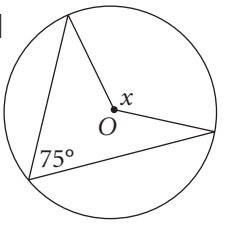
6 

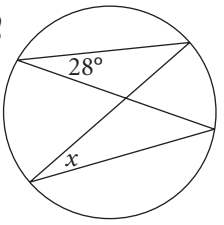
7 

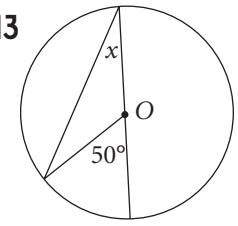
8 

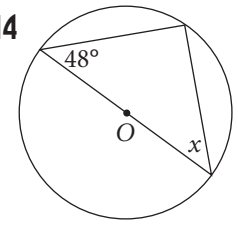
9 

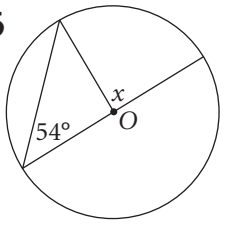
10 

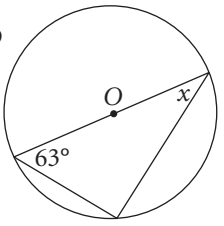
11 

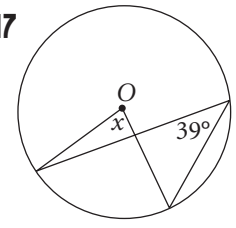
12 

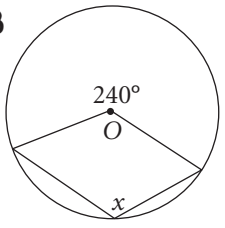
13 

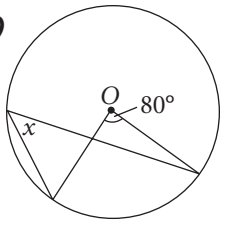
14 

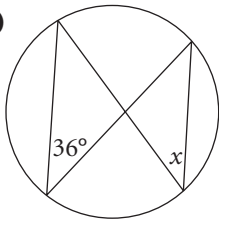
15 

16 

17 

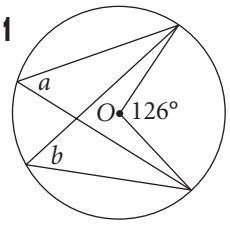
18 

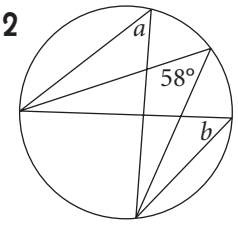
19 

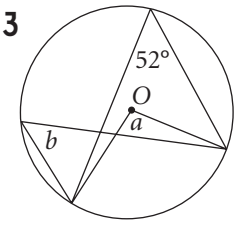
20 

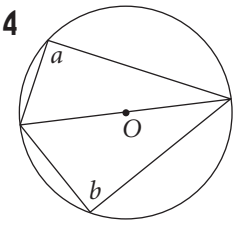
C

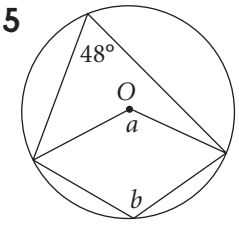
Find the value of each pronumeral in the following diagrams. (*O* is the centre of each circle.)

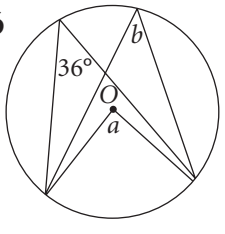
1 

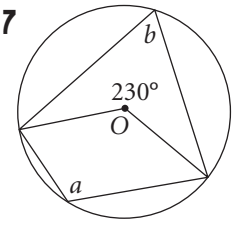
2 

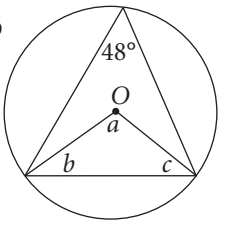
3 

4 

5 

6 

7 

8 

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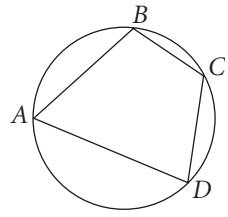
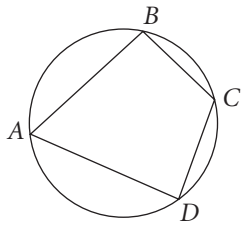
Cyclic quadrilaterals 15.1

If a circle can be drawn so that it passes through all four vertices of a quadrilateral, then the figure is called a *cyclic quadrilateral.*



A Investigation

1 Construct a circle on a sheet of paper, using a radius of 5 cm or more. Mark any four points *A*, *B*, *C* and *D* on the circumference of this circle. Join then to form a cyclic quadrilateral *ABCD.*



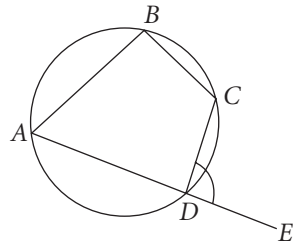
2 Use a protractor to measure the size of $\angle BAD$ and $\angle BCD$. What is the sum of these two angles? (Add $\angle BAD$ and $\angle BCD$.)

3 Now measure $\angle ABC$ and $\angle ADC$. What is the sum of $\angle ABC$ and $\angle ADC$?

4 Repeat 1, 2 and 3 with another circle and different positions for the points *A*, *B*, *C* and *D*.

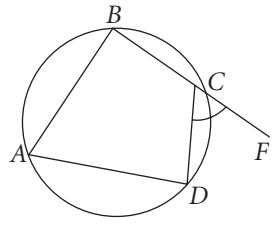
5 Do the opposite angles of a cyclic quadrilateral always add to give 180°?

6 Using your diagram from 1, produce the side *AD* to a point *E*, forming an exterior angle *CDE*, as shown.



7 Measure the size of $\angle CDE$. Does $\angle CDE = \angle ABC$?

8 Produce the side *BC* to a point *F*, forming an exterior angle *DCF.*



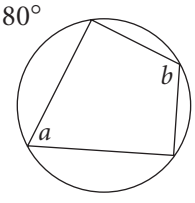
9 Measure the size of $\angle DCF$. Does $\angle DCF = \angle BAD$?

10 Repeat 6 to 9 using a second cyclic quadrilateral.

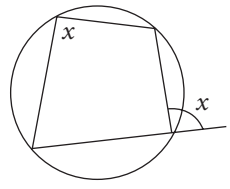
11 Does the exterior angle of a cyclic quadrilateral always equal the opposite interior angle?

You have demonstrated the following properties of a cyclic quadrilateral:

- The opposite angles of a cyclic quadrilateral add to give 180° ($a + b = 180°$).

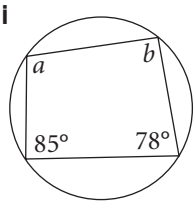


- An exterior angle of a cyclic quadrilateral is equal to the opposite interior angle.



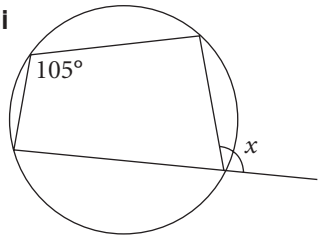
Example: Find the value of each pronumeral.

i



$a = 180° - 78°$
$= 102°$
$b = 180° - 85°$
$= 95°$

ii



$x = 105°$

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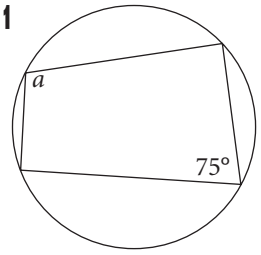
B Find the value of each pronumeral in the following diagrams.

1



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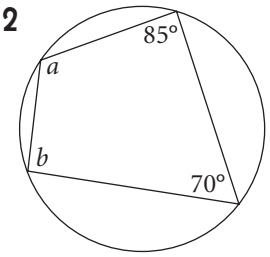

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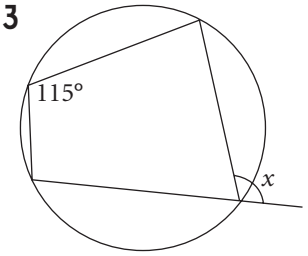

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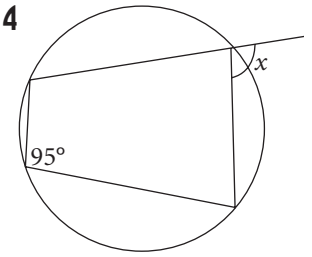

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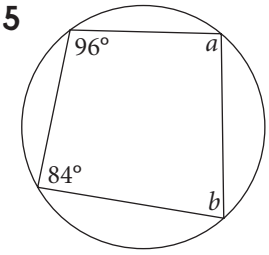

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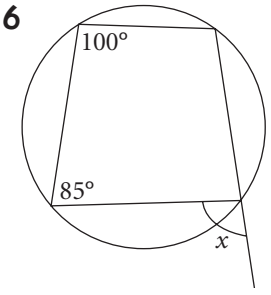

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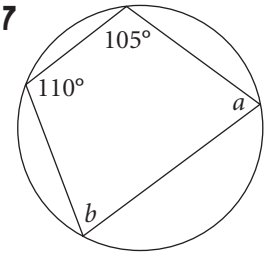

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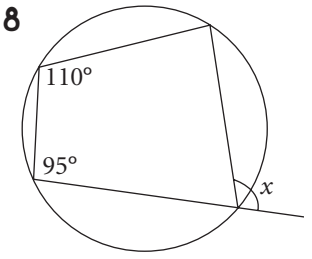

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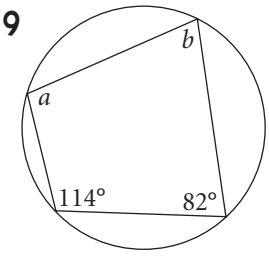

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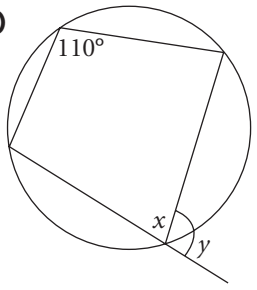

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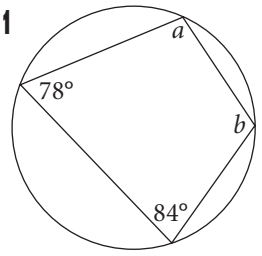

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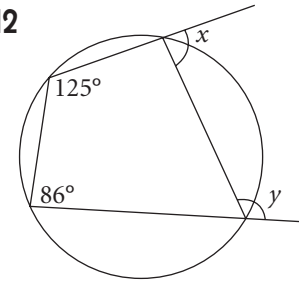

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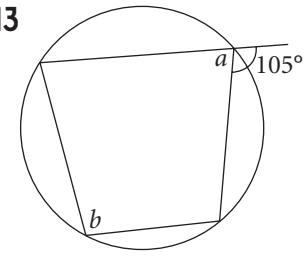

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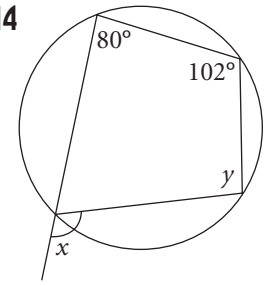

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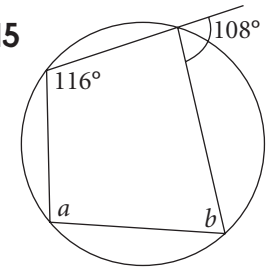

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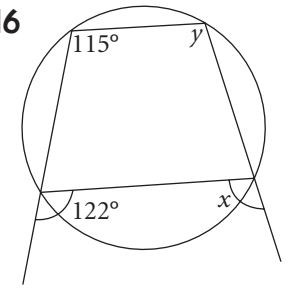
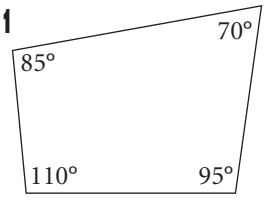


C Is it possible to draw a circle passing through all four vertices of these quadrilaterals?
(*Hint:* In a cyclic quadrilateral, each pair of opposite angles add to 180°.)

1



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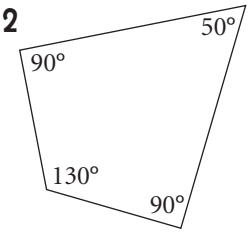

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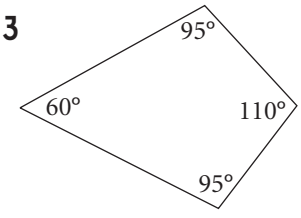

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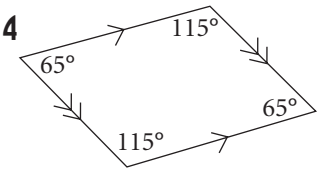

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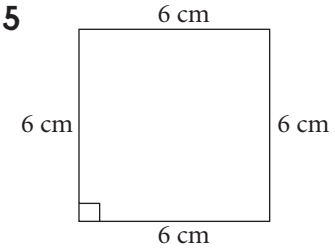

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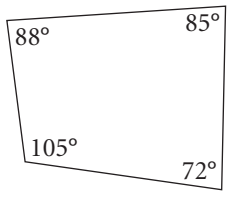

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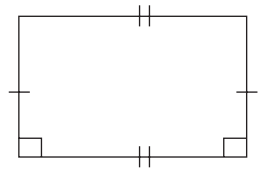

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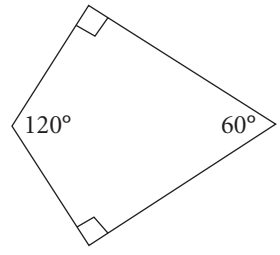

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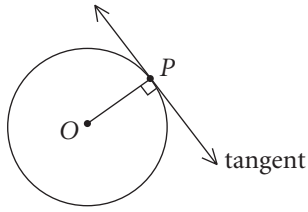
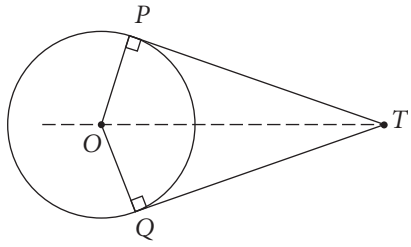
Tangents to circles 16.1

A





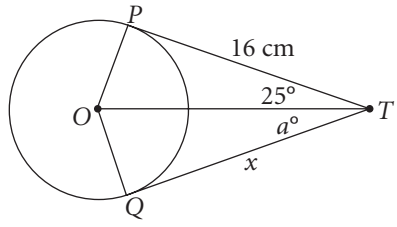
- A *tangent* is a straight line which touches a circle at only one point. This point (P) is called the point of contact.
- The tangent meets the radius drawn to the point of contact at right angles.
- From any external point (T), two tangents of equal length may be drawn to a circle ($PT = QT$).
- The line joining the external point (T) to the centre of the circle is an axis of symmetry of this diagram.

Example:

PT and QT are tangents to the circle, centre O.
$PT = 16$ cm and $\angle OTP = 25°$.
Find:

a the length of QT

b the size of $\angle OTQ$

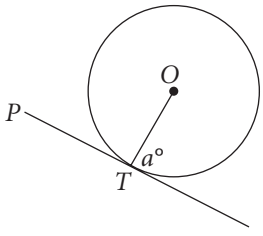


Solution:

$QT = PT$ (equal tangents from an external point)
$= 16$ cm
$\angle OTQ = \angle OTP$ (the line OT is an axis of symmetry)
$= 25°$

In the following diagrams, PT and QT are tangents drawn to the circle, centre O. Find the value of each pronumeral.

1



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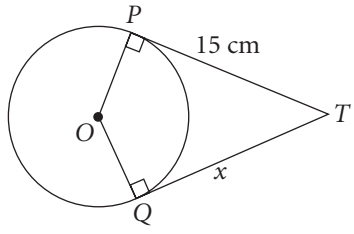

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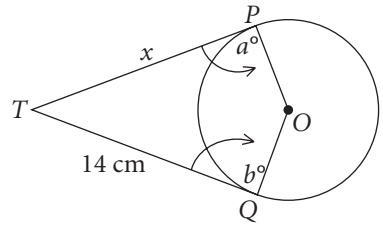

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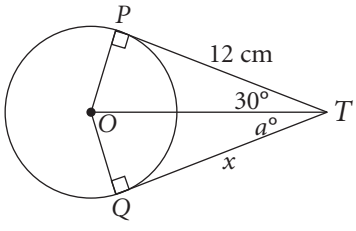

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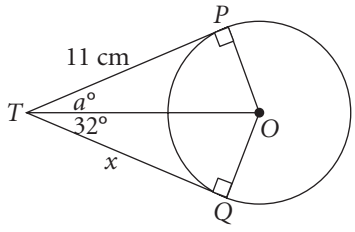

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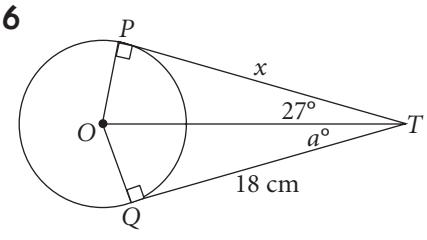

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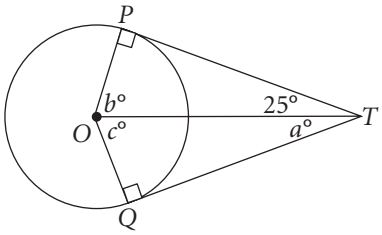

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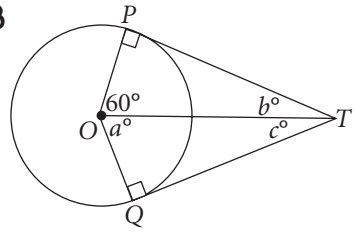

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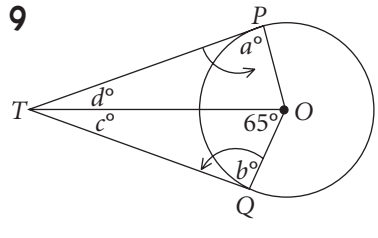

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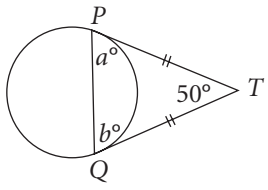
B

In the following diagrams, PT and QT are tangents drawn to the circle.
Use the properties of isosceles triangles to find the value of each pronumeral.

1



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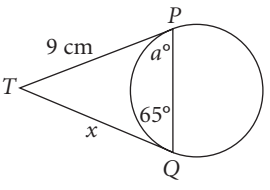

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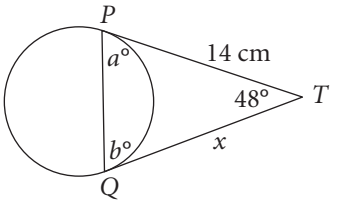

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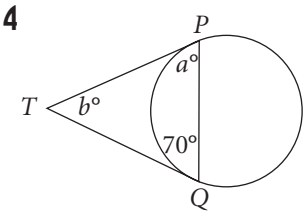

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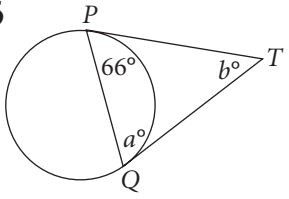

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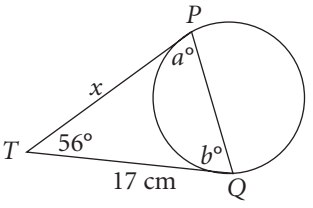
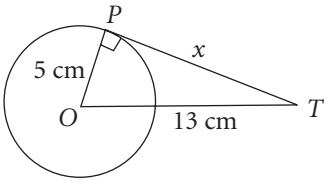


Example:

PT is a tangent to the circle, centre O.
Use Pythagoras' theorem to calculate the length of the tangent PT.



Solution:

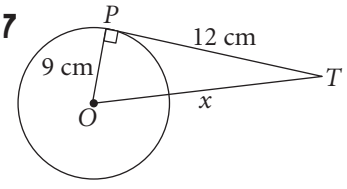
Applying Pythagoras' theorem:

$$OT^2 = OP^2 + PT^2$$
$$13^2 = 5^2 + x^2$$

Rearranging:
$$x^2 = 13^2 - 5^2$$
$$x = \sqrt{144}$$
$$= 12 \text{ cm}$$

Use Pythagoras' theorem to calculate the distance marked x in each diagram. (O is the centre.)

7



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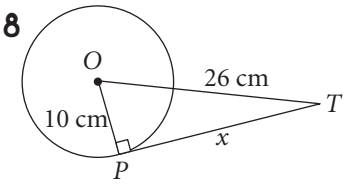

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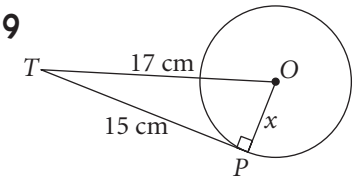

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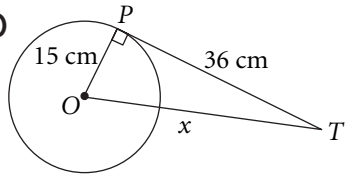
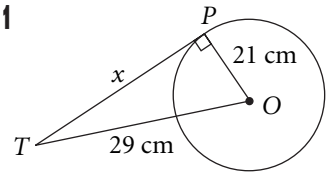


11



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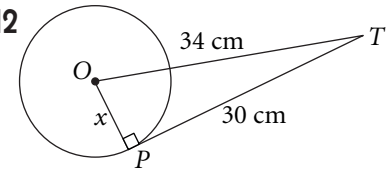
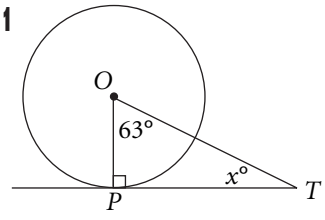


C

Find the value of each pronumeral.(PT and QT are tangents, O is the centre of the circle.)

1



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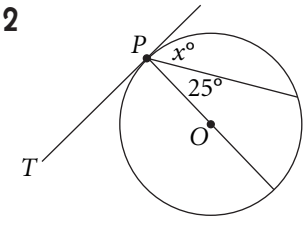

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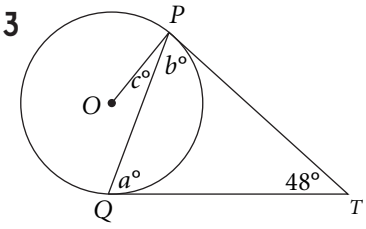

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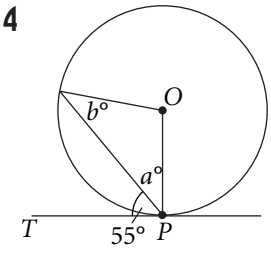

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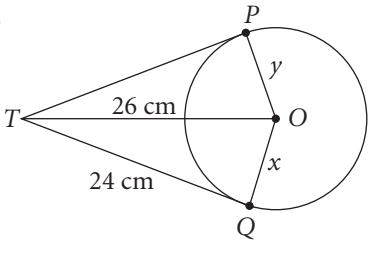

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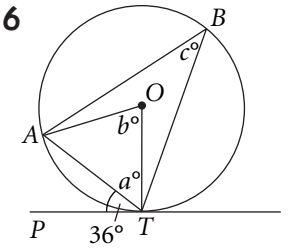

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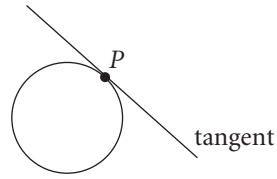
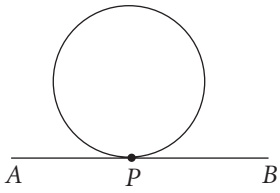
The angle in the alternate segment 17.1

A tangent is a straight line which touches a circle at one point only.
This point (P) is called the point of contact.

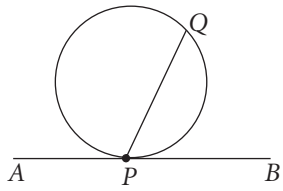


A Investigation

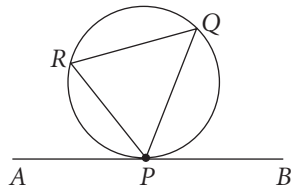
1 Construct a circle on a sheet of paper (using a radius of 5 cm or more) and draw a tangent to the circle. Label the point of contact (P) and two points on the tangent A and B, as shown.



2 Draw a chord in the circle using P as one endpoint, and label the other endpoint Q.



3 Draw a second chord PR, then join the points R and Q.



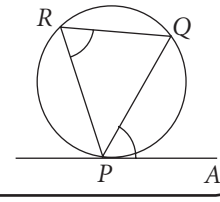
4 Use a protractor to measure the size of $\angle BPQ$ and $\angle PRQ$. What do you notice?

5 Measure the size of $\angle APR$ and $\angle PQR$. What do you notice?

6 Repeat 1 to 5 using a circle with a different radius, and change the position of the point Q on the circumference of the circle.

You have demonstrated the following angle property of a circle:

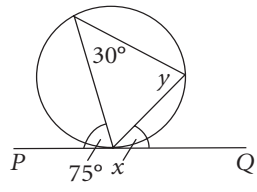
> The angle between a tangent and a chord, drawn to the point of contact, is equal to the angle in the alternate segment (standing on this chord).
> $\angle APQ = \angle PRQ$



B

Example:

PQ is a tangent to a circle.
Find the size of the angles marked x and y.

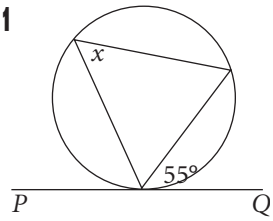


Solution:

$x = 30°$
$y = 75°$

Find the value of the pronumeral x in each diagram. (PQ is a tangent).

1



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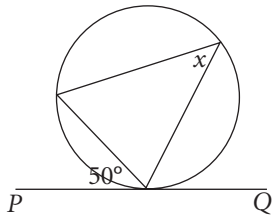

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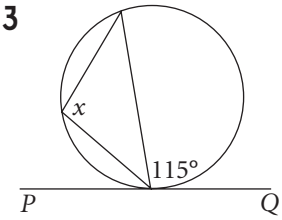

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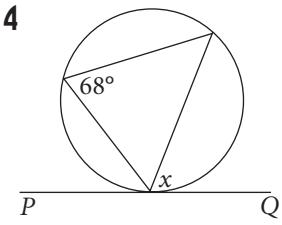

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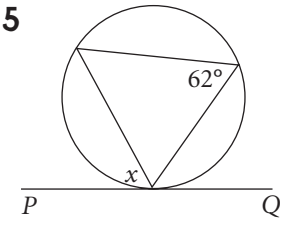

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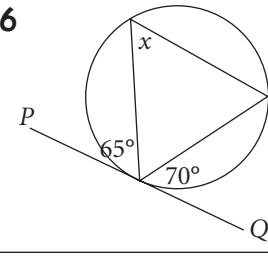

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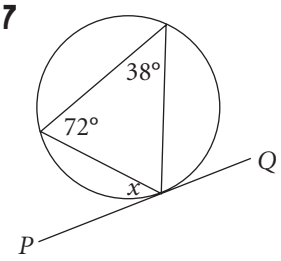

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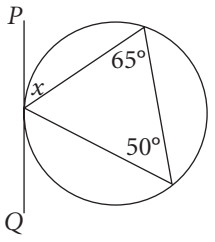

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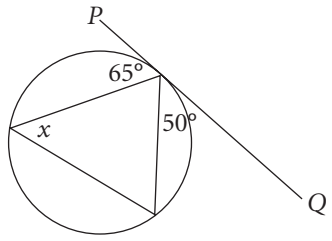

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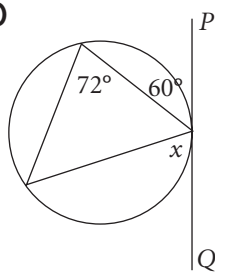

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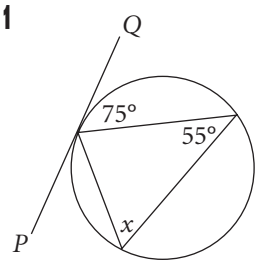

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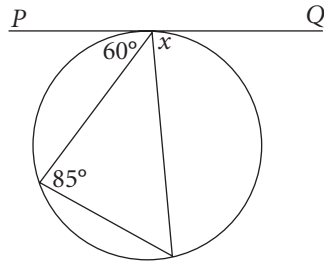

C

Find the values of the pronumerals in each diagram. (*PQ*, *PR* and *TS* are tangents.)

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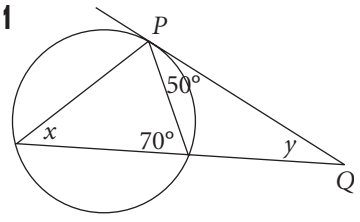

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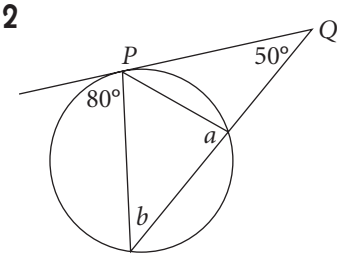

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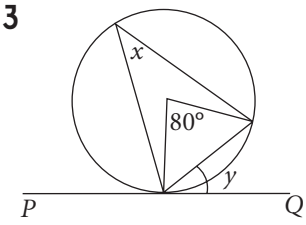

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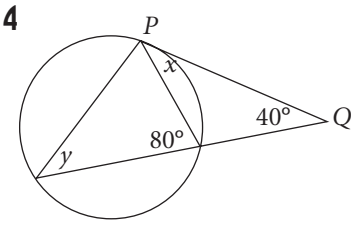

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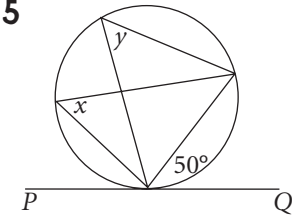

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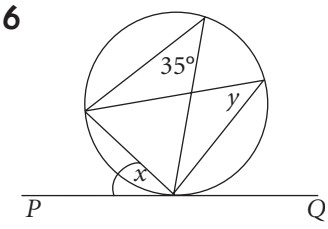

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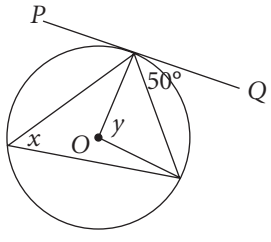

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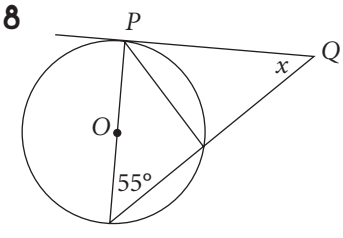

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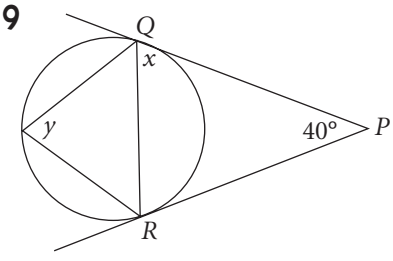

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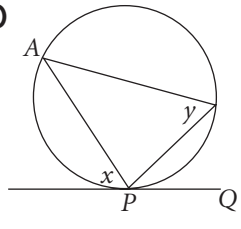
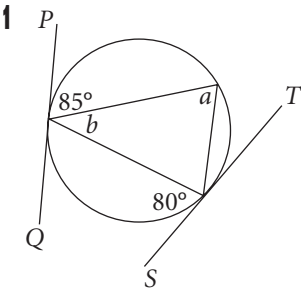


$\angle APQ = 125°$

11



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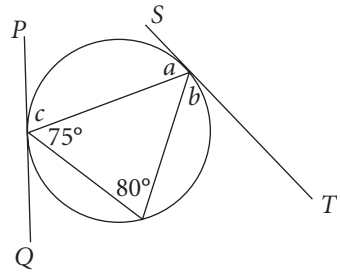

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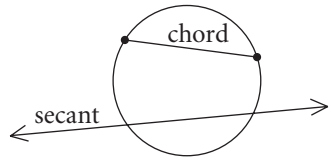
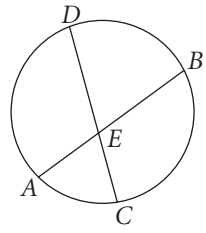
Intersecting chords and secants 18.1

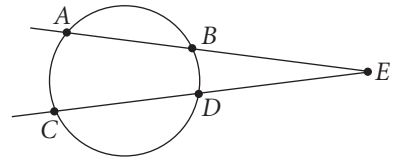
A *chord* is an interval joining two points on a circle.
A *secant* is a straight line that cuts a circle at two distinct points.



A Investigation



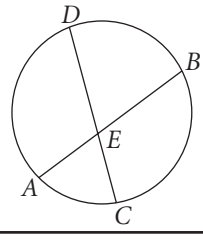
1. Construct a circle an a sheet of paper, using a radius of 4 cm or more.
2. Use a ruler to draw two intersecting chords *AB* and *CD* and label the point of intersection *E*.
3. Use a ruler to measure the lengths of the intervals *AE*, *EB*, *CE* and *ED* to the nearest millimetre.
4. Are the products $AE \times EB$ and $CE \times ED$ equal?
5. Repeat 1, 2 and 3 with another circle and change the positions of the points *A*, *B*, *C* and *D*. Are the products of the intercepts $AE \times EB$ and $CE \times ED$ equal?



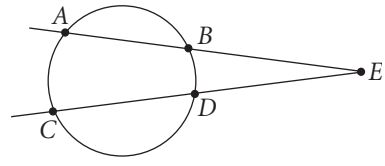
6. Construct another circle on your sheet of paper, and use a ruler to draw two *secants* that meet outside the circle. Label the points *A*, *B*, *C*, *D* and *E* as shown.
7. Use your ruler to measure the lengths of the intervals *AE*, *EB*, *CE* and *ED* to the nearest millimetre.
8. Are the products $AE \times EB$ and $CE \times ED$ equal?

You have demonstrated two properties of chords and secants:

- The products of the intercepts of two *intersecting chords* of a circle are equal.
  $AE \times EB = CE \times ED$



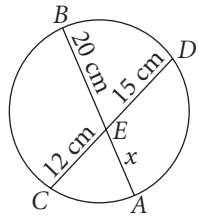
- The products of the intercepts of two *intersecting secants* to a circle from an external point are equal.
  $AE \times EB = CE \times ED$



B

Example:

The chords *AB* and *CD* intersect at the point *E*.
Find the value of *x*.

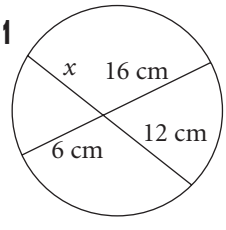


Solution:

$AE \times EB = CE \times ED$
$\therefore\ x \times 20 = 12 \times 15$
$20x = 180$
$x = 9\text{cm}$

Find the value of *x* in each diagram.

1



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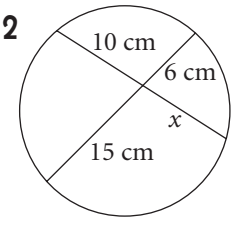

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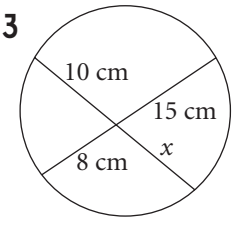
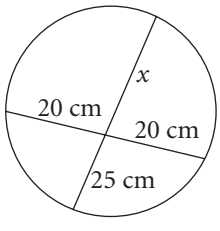
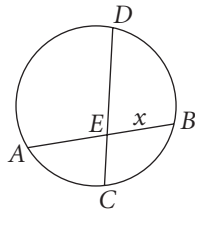


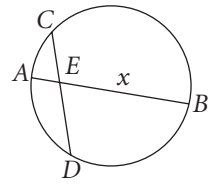
4



5 $AE = 16\text{ cm}$
$CE = 10\text{ cm}$
$ED = 20\text{ cm}$
$EB = x$



6 $AE = 5\text{ cm}$
$CE = 8\text{ cm}$
$ED = 10\text{ cm}$
$EB = x$



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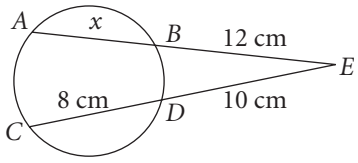
Example:

The secants AB and CD intersect at the point E. Find the value of x.

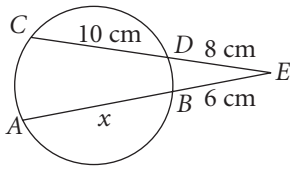


Solution:

$$AE \times EB = CE \times ED$$
$$(x + 12) \times 12 = 18 \times 10$$
$$12x + 144 = 180$$
$$12x = 36$$
$$x = 3\text{ cm}$$

Find the value of x in each diagram.

7



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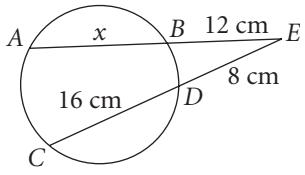
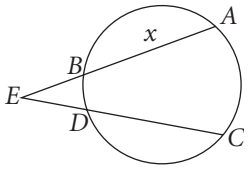
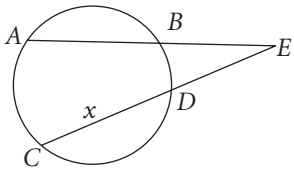


9 $EB = 8\text{ cm}$
$CD = 12\text{ cm}$
$DE = 12\text{ cm}$
$AB = x$

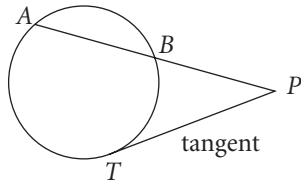


10 $AB = 9\text{ cm}$
$BE = 9\text{ cm}$
$ED = 10\text{ cm}$
$CD = x$



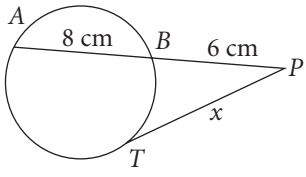
C

- A *tangent* to a circle is a straight line that touches the circle at one point only.
- The square of the tangent to a circle from an external point is equal to the product of the intercepts of any secant from that point.
  $PT^2 = AP \times PB$



Example:

The tangent PT and the secant AB meet at the point P. Find the length of the tangent PT.

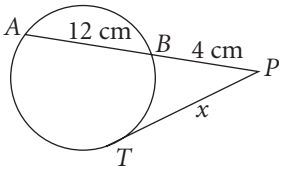


Solution:

$$PT^2 = AP \times PB$$
$$\therefore x^2 = 14 \times 6 \quad (AP = 8\text{ cm} + 6\text{ cm})$$
$$= 84$$
$$x = \sqrt{84}$$
$$= 9.2\text{ cm (to 1 decimal place)}$$

Calculate the length of the tangent PT in each diagram, correct to 1 decimal place, where necessary.

1



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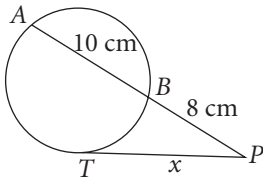

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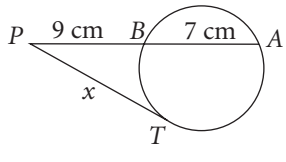

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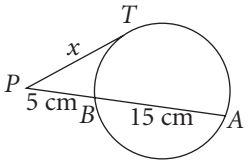

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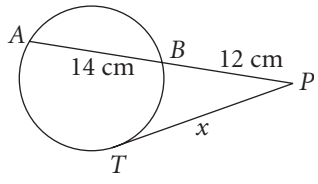

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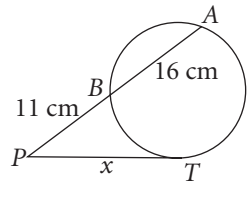

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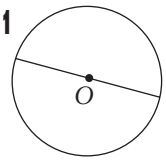
Review — Properties of circles 19.1

A Name the feature or part of a circle indicated in each diagram.

1



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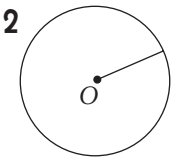

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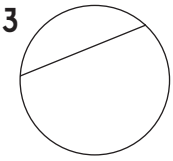

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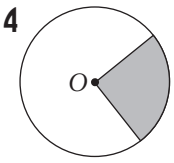

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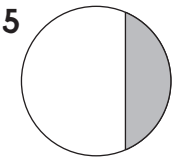

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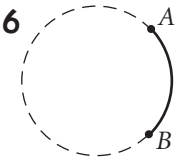

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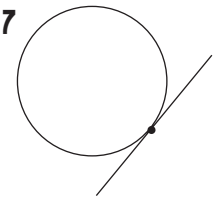

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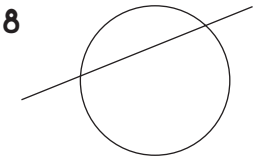


Complete these statements of some of the properties of circles using the wordbank to fill in the missing words.

Wordbank: *angles, arc, centre, chords, circle, circumference, distance, equal, opposite, perpendicular, quadrilateral, standing, subtended, through, twice.* (Each word may be used once only.)

9 Equal _____ are the same _____ from the centre of the circle.

10 Equal chords subtend equal _____ at the centre of the _____.

11 The _____ bisector of a chord passes _____ the _____ of the circle.

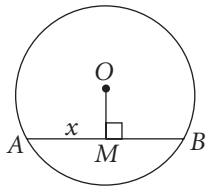
12 Angles at the _____ of a circle, _____ on the same arc, are _____ in size.

13 The angle subtended by an _____ at the centre of a circle is _____ the size of an angle _____ by the same arc at the circumference.

14 The _____ angles of a cyclic _____ add to 180°.

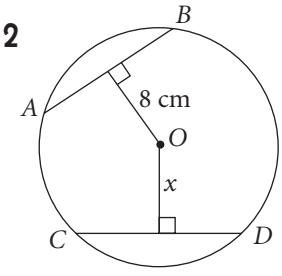
B Find the values of the pronumeral in each diagram. (*O* is the centre of each circle.)

1



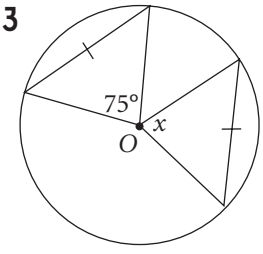
AB = 18 cm, find the length of *AM*

2



AB = *CD* = 12 cm

3



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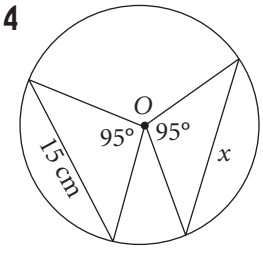

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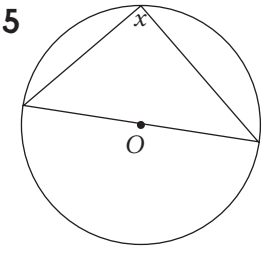

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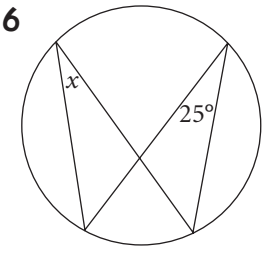

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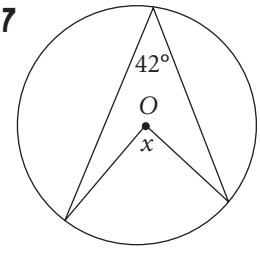

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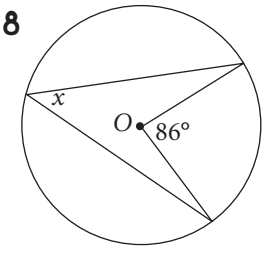

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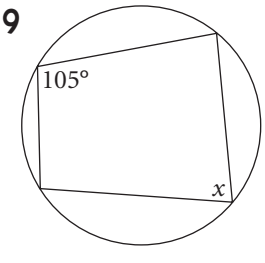

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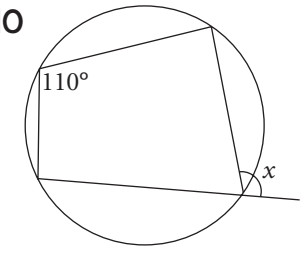

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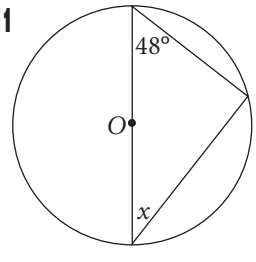

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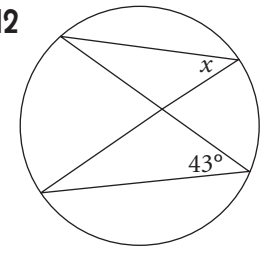

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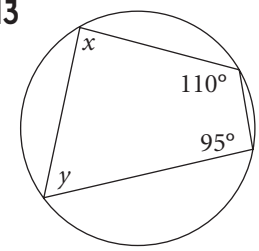
13



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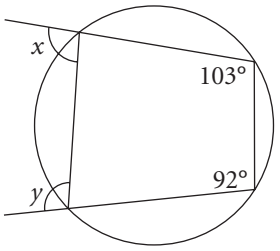

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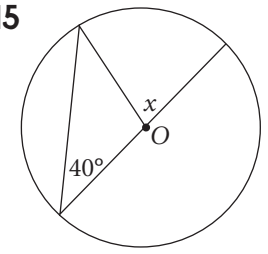

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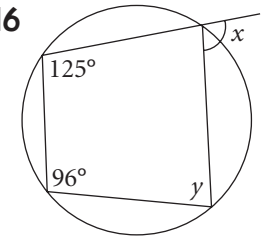
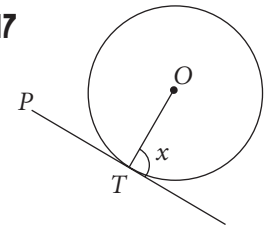


In the following diagrams *PT* and *PQ* are tangents to the circles.

17



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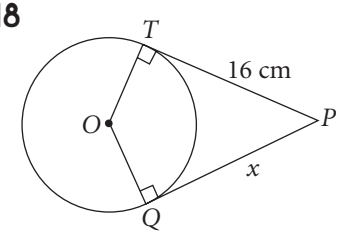

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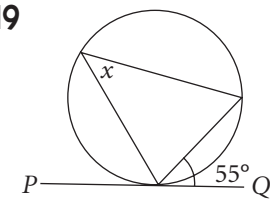

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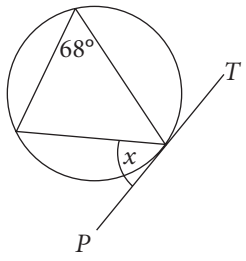
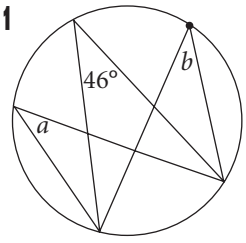


C

Find the values of the pronumerals in each diagram. (*O* is the centre of each circle.)

1



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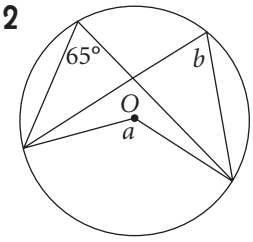

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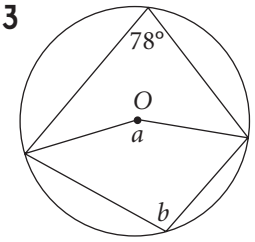

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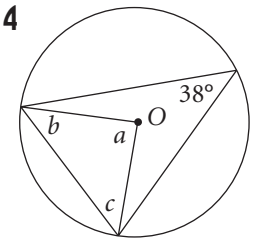
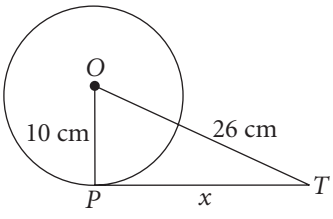


PT and *QT* are tangents drawn to each circle.

5



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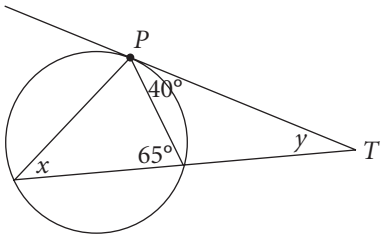

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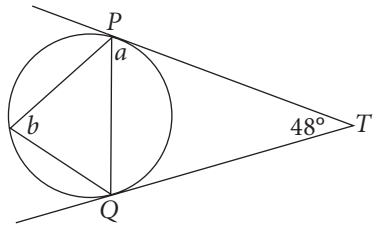
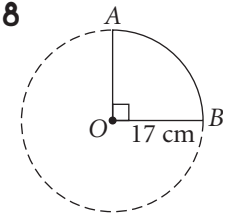


Calculate the length of the arc *AB* shown in each diagram, correct to one decimal place.

8



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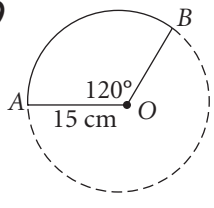

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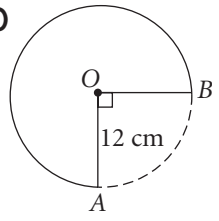

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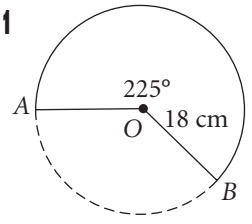
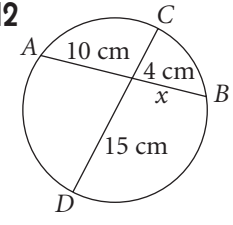
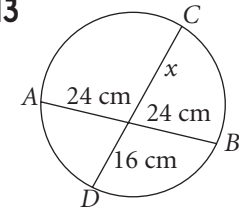


Find the value of *x* in each diagram, correct to 1 decimal place where necessary.

12



13



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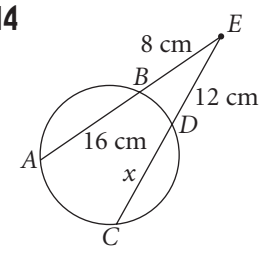

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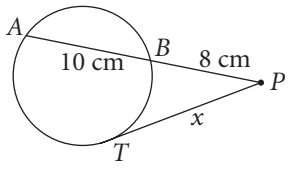

PT is a tangent

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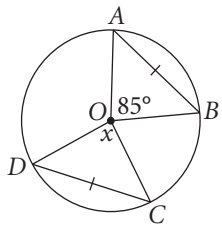
NOTE: DIAGRAMS NOT TO SCALE

Circles and deductive reasoning 20.1

A



Example:

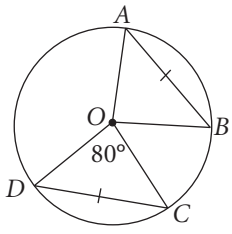
AB and *CD* are equal chords of the circle centre *O*. $\angle AOB = 85°$.
Find the size of $\angle DOC$, giving a brief reason for your answer.

Solution:

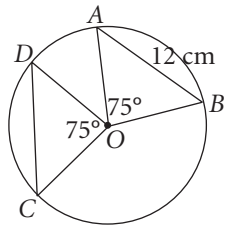
$\angle DOC = 85°$ (Equal chords subtend equal angles at the centre of the circle.)

In each question give a *brief reason* for your answer. (*O* is the centre of each circle.)

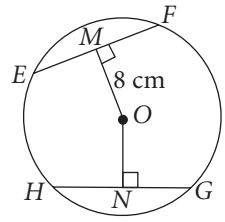
1 $AB = CD$
$\angle DOC = 80°$
Find the size of $\angle AOB$.



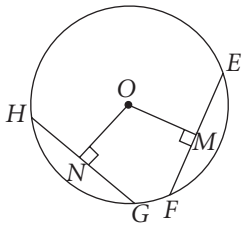
2 $\angle AOB = \angle DOC$
$AB = 12$ cm
Find the length of *CD*.



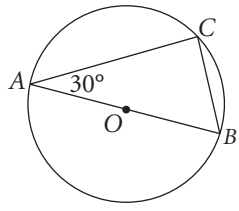
3 $EF = GH$
$OM = 8$ cm
Find the length of *ON*.



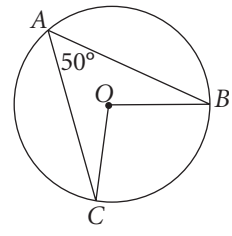
4 $ON = OM$
$EF = 10$ cm
Find the length of *GH*.



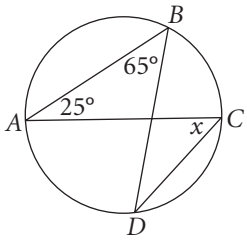
5 *AB* is a diameter of the circle.
Find the size of $\angle ACB$.



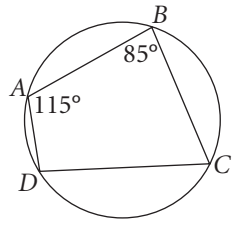
6 $\angle BAC = 50°$
Find the size of $\angle BOC$.



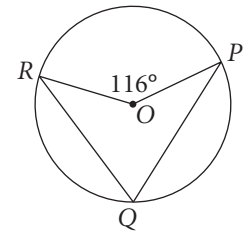
7 Find the size of $\angle ACD$.



8 Find the size of $\angle BCD$.



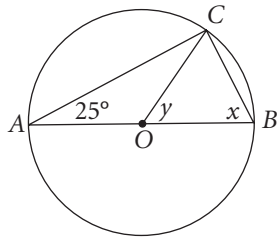
9 $\angle ROP = 116°$
Find the size of $\angle RQP$.



B

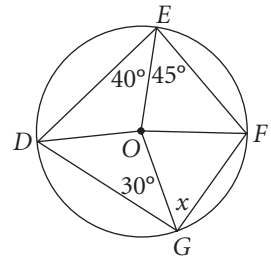
In each diagram, *O* is the centre of the circle.

1 Find the sizes of the angles marked *x* and *y*, giving brief reasons for each answer.



2 **a** How many isosceles triangles are there in this diagram?

b Find the size of the angle *x* giving brief reasons for your answer.



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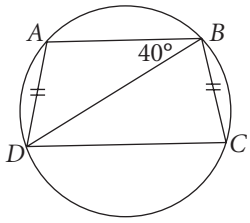
3 Given $BC = AD$ and $\angle ABD = 40°$,

a find the size of $\angle BDC$ (give reasons)

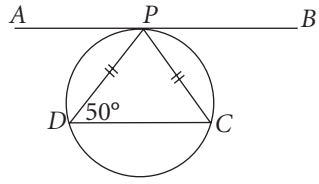
b explain why $AB \parallel DC$.



4 AB is a tangent to the circle at the point P. $DP = CP$ and $\angle PDC = 50°$.

a Find the size of $\angle BPC$ giving brief reasons.

b Explain why $AB \parallel DC$.



5 AB is a chord of the circle centre O. M is the midpoint of the chord.

a Complete this proof:

In the Δs AMO and BMO,

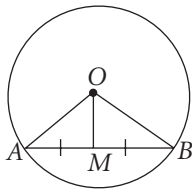
$AM =$ _____ (given)

$OA =$ _____ (radii of the circle)

OM is _____

$\therefore \Delta AMO \equiv \Delta BMO$ (_____ test)

b Explain why OM is perpendicular to the chord AB.



6 The chords AB and CD subtend equal angles at the centre of this circle.

a Complete this proof:

In the Δs AOB and DOC,

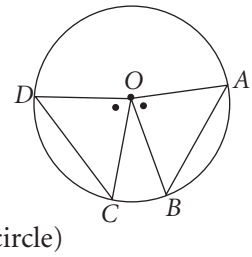
$OA =$ _____ (radii of the circle)

$\angle AOB =$ _____ (_____)

$OB =$ _____ (_____)

$\therefore \Delta AOB \equiv \Delta DOC$ (_____ test)

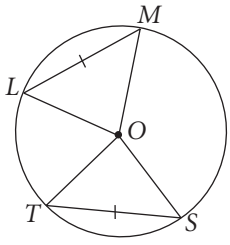
b Explain why $AB = CD$.



7 The chords LM and ST are the same length.

a Prove that $\Delta LOM \equiv \Delta SOT$.

b Explain why $\angle LOM = \angle SOT$.



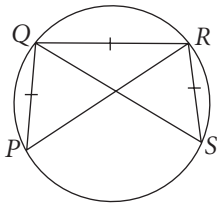
8 The chords PQ, QR and RS are the same length.

a Explain why $\angle QPR = \angle QSR$.

b Explain why $\angle PRQ = \angle SQR$.

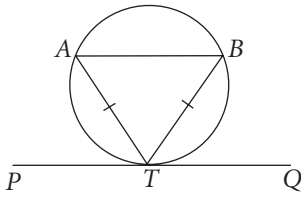
c Prove that $\Delta PQR \equiv \Delta SRQ$.

d Explain why $PR = QS$.



C

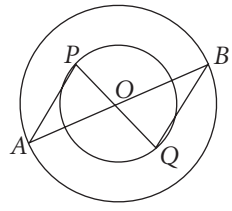
1 PQ is a tangent to the circle at the point T and $AT = BT$. Prove that $AB \parallel PQ$.



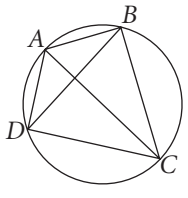
2 The point O is the centre of both circles. AB is a chord of the larger circle. PQ is a chord of the smaller circle.

a Prove that $\Delta AOP \equiv \Delta BOQ$.

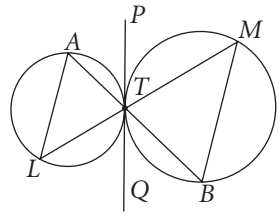
b Explain why $AP \parallel QB$.



3 $ABCD$ is a cyclic quadrilateral. AC bisects $\angle BCD$. Prove that ΔABD is isosceles.



4 PQ is a common tangent to the two circles which touch at the point T. AB and LM are straight lines. Prove that $AL \parallel MB$.



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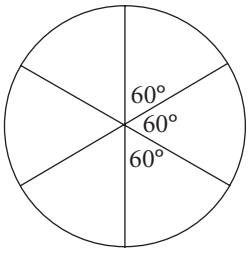
NOTE: DIAGRAMS NOT TO SCALE

Hexagon designs 21.1

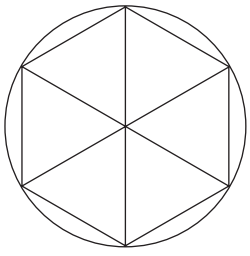
Many interesting designs are constructed from a *hexagon* inscribed in a circle. These constructions will be much easier when you use a circle with a radius of 5 cm or more.

Use a pencil for all construction lines. You may want to erase some of them later.

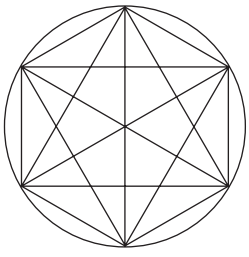
1 **a** Construct a circle and draw 3 diameters at angles of 60° to each other.



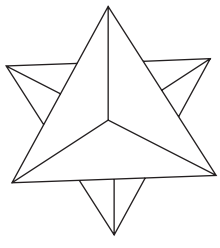
b Join the six points around the circle to form a hexagon.

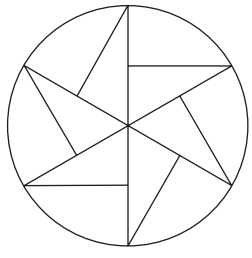


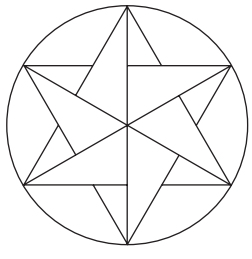
c Draw the remaining diagonals of the hexagon. (Join each vertex to every other vertex.)



d By carefully erasing some construction lines, can you obtain each of these designs?

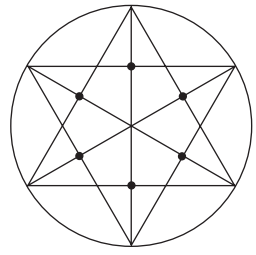




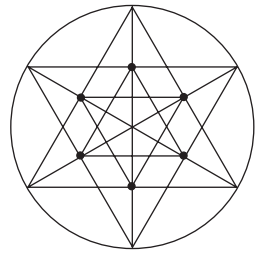


2 **a** Construct a circle and draw 3 diameters at angles of 60° to each other.

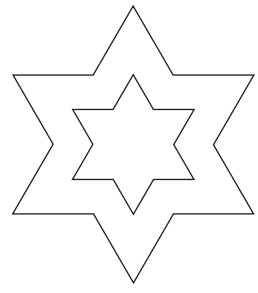
b Join each point on the circle to the *second* point away from it.

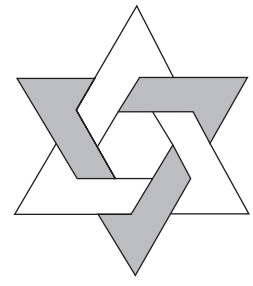


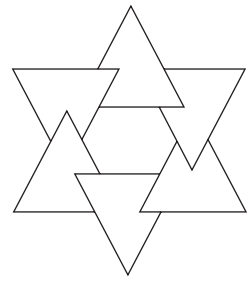
c Add the additional construction lines shown in this diagram.
(Each of the points marked with the bold dot was joined to the second point away from it.)



d By carefully erasing some construction lines, can you obtain each of these designs?







They will look even better if you add colour!

NOTE: DIAGRAMS NOT TO SCALE

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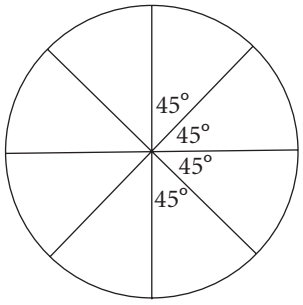
Octagon designs 22.1

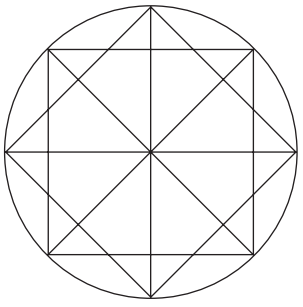
Many designs are based on an *octagon* inscribed in a circle. These constructions will be much easier when you use a circle with a radius of 5 cm or more.

Use a pencil for all construction lines. You may want to erase some of them later.

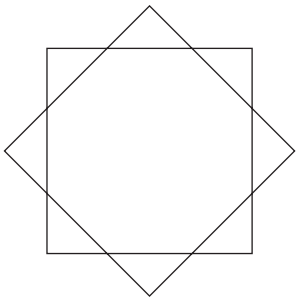
1 **a** Construct a circle and draw 4 diameters at angles of 45° to each other.

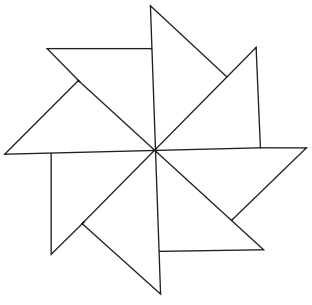


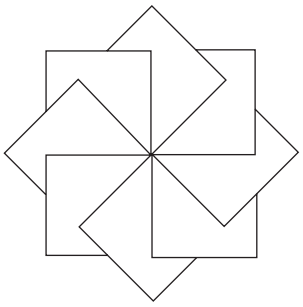
b Join each point on the circle to the second point from it, working clockwise.



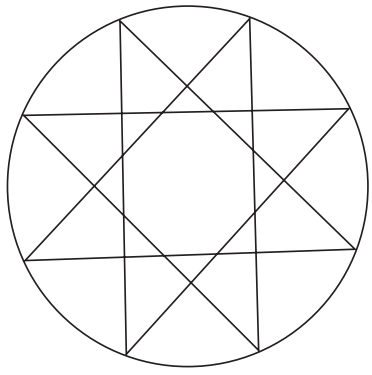
c By carefully erasing the circle and some of the construction lines, can you obtain each of these designs?

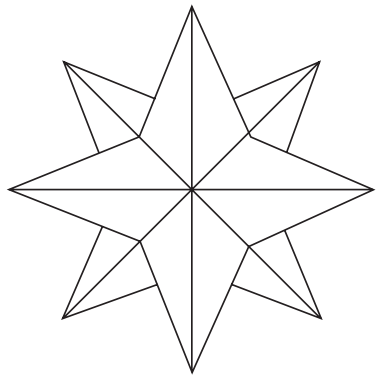


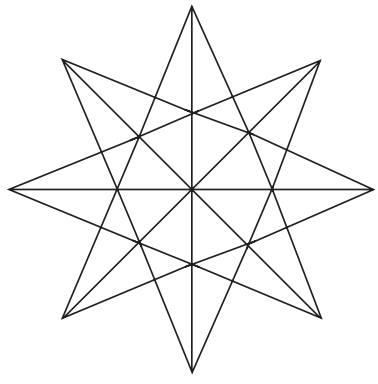




2 The following designs begin with a circle divided into 8 sectors. Can you work out how to construct them? They will look even better if you add colour.







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1—Geometric constructions

A **5** **d** They meet at a common point.

3—Constructing quadrilaterals

A **1** **a** rectangle **b** parallelogram **c** rhombus
d square **e** trapezium **f** rhombus
g kite **h** trapezium

C **1** **c** $AB = 45$ mm, $BC = 28$ mm, $CD = 28$ mm, $AD = 45$ mm
d kite
2 **c** All sides measure 36 mm. **d** rhombus

4—Constructions and circles

A **2** **a** yes **b** no **c** no **d** no

C **3** **a** 72° **b** 54° **5** **a** 45°

5—Similar figures

A **1** *A–B, C–H, D–N, E–P, F–L, G–I, J–Q, K–O, M–R*
2 **a** The length of rectangle B is *two* times the length of rectangle A. The breadth of rectangle B is *two* times the length of rectangle A. **b** 1 : 2
3 **a** yes **b** true

B A : H = 1 : 2, B : F = 5 : 4, C : I = 5 : 8, D : E = 1 : 2, G : J = 5 : 4

C **1** 8 cm **2** 9 cm **3** 6 cm **4** 18 cm **5** 9 cm
6 12 cm

7—Similar triangles

A **1** $\angle A = \angle D, \angle B = \angle E, \angle C = \angle F$
3a 0.5 **3b** 0.5 **3c** 0.5 **4** 1 : 2

B **1** yes **2** no **3** yes
4 A–L, B–N, C–K, D–O, E–P, F–Q, G–R, H–M

C **1** In the Δs *ABC* and *ADE*,
$\angle BAC$ is common
$\angle ABC = \angle ADE$
$\angle ACB = \angle AED$ (equal corresponding angles, $BC \parallel DE$)
$\therefore \Delta ABC \,|||\, \Delta ADE$ (They are equi-angular.)

2 In the Δs *PRT* and *QST*,

$\frac{PT}{QT} = \frac{6}{9} = \frac{2}{3}$ and $\frac{RT}{ST} = \frac{4}{6} = \frac{2}{3}$

$\angle PTR = \angle QTS$ (vertically opposite angles)
$\therefore \Delta PRT \,|||\, \Delta QST$ (Two pairs of corresponding sides are in the same ratio and the included angles are equal.)

8—Using similar triangles

A **1** A–D, B–C, E–G, F–H **2** $\frac{AB}{MN} = \frac{BC}{LN} = \frac{AC}{LM}$

B **1** $x = 12$ cm **2** $x = 6$ cm, $y = 4$ cm
3 $x = 16$ cm, $y = 20$ cm **4** $y = 12$ cm
5 $x = 36$ cm, $y = 45$ cm **6** $x = 20$ cm, $y = 18$ cm
7 $x = 15$ cm, $y = 26$ cm **8** $x = 5.4$ cm, $y = 12.5$ cm
9 $x = 12$ cm **10** $x = 14.4$ cm, $y = 25$ cm

C **1** $x = 10$ cm **2** $x = 22.5$ cm **3** $x = 9$ cm
4 $x = 12.5$ cm, $y = 16.25$ cm **5** 6 m
6 1.05 m or 105 cm

9—Similar triangles and reasoning

A **1** yes **2** yes **3** no **4** yes **5** no **6** yes

B **1** $\Delta ABE \,|||\, \Delta DCE$ (they are equiangular, $\angle A = \angle D$, $\angle B = \angle C$)
2 $\Delta ABE \,|||\, \Delta ACD$ ($\frac{AB}{AC} = \frac{AE}{AD} = \frac{5}{9}$ $\angle A$ is common)
3 $\Delta ADE \,|||\, \Delta CBE$ (matching sides are in the ratio 3:2)
4 **b** 30 cm
5 **b** 26 cm
6 **b** 6.4 cm
7 **b** 21 cm
8 **b** 5.25 cm
9 **b** 15 cm

C **4** **b** No, the pairs of alternate angles are *not* equal
8 **b** 1 : 2

10—Areas of similar figures

A **1** **a** 3 : 4 **b** A = 9 cm^2, B = 16 cm^2 **c** 9 : 16
2 **a** 1 : 2 **b** 1 : 2 **c** X = 12 cm^2, Y = 48 cm^2
d 1 : 4
3 **a** 3 : 2 **b** 3 : 2 **c** A = 54 cm^2, B = 24 cm^2 **d** 9 : 4

B **1** **a** 1 : 2 **b** 1 : 4
2 **a** Corresponding sides are in the same ratio. **b** 1 : 2
c 1 : 4
3 9 : 25 **4** **a** 3 : 4 **b** 9 : 16 **5** 64cm^2
6 25.2 cm^2 **7** 100 cm^2 **8** 40 cm^2 **9** 4
10 **a** 16 : 9 **b** 4 : 3 **11** 5 : 3 **12** 22.5 cm

C **1** **a** 1 : 2 **b** 16 cm^2, 96 cm^2 **c** 64 cm^2, 384 cm^2
d 1 : 4 **e** 1 : 4 **f** A = 64 cm^3, B = 512 cm^3 **g** 1 : 8
2 **a** 1 : 3 **b** 108 cm^2 **c** 972 cm^2 **d** 1 : 9
e 72 cm^3 **f** 1944 cm^3 **g** 1 : 27
3 **a** If the lengths of corresponding edges are in the ratio a : b, then the surface areas are in the ratio a^2 : b^2.
b If the lengths of corresponding edges are in the ratio a : b then the volumes are in the ratio a^3 : b^3.

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11—Review — Similar figures

A **1** A–D, B–E, C–I, F–G, H–J
2 B and C **3** **a** yes **b** no **c** no **d** yes

B **1** $x = 60$ cm **2** $x = 18$ cm **3** $x = 18$ cm
4 $x = 18$ cm, $y = 12$ cm **5** $x = 12$ cm
6 $x = 6$ cm, $y = 10$ cm **7** $x = 30$ cm, $y = 54$ cm
8 $x = 16$ cm, $y = 15$ cm **9** $x = 14.4$ cm, $y = 25$ cm
10 $x = 12$ cm **11** 9 : 16 **12** 324 cm^2

C **1** In the Δs ABD and ECD,
$\angle ADB$ is common
$\angle DAB = \angle DEC$ (equal corresponding angles, $AB \parallel EC$)
$\angle ABD = \angle ECD$ (equal corresponding angles, $AB \parallel EC$)
$\therefore \Delta ABD \,|||\, \Delta ECD$ (they are equiangular)

2 **b** Pairs of alternate angles are equal. ($\angle P = \angle Q$, $\angle R = \angle S$)
3 **b** 20 cm

12—Circles

B **1** radius **2** chord **3** diameter **4** minor segment
5 sector **6** major segment **7** semicircle
8 arc LM **9** chord **10** tangent
11 secant **12** semicircle **13** minor segment
14 arc PQ **15** tangent **16** sector

C **1** 23.6 cm **2** 17.8 cm **3** 37.7 cm **4** 66.0 cm
5 31.4 cm **6** 37.7 cm **7** 47.1 cm **8** 41.9 cm

13—Chord properties of circles

A **1** $PQ = RS$, $OC = OD$ **2** $KL = NM$, $OX = OY$
3 $CD = FE$, $\angle DOC = \angle EOF$, $OC = OF$, $OD = OE$
4 $PQ = SR$, $\angle POQ = \angle SOR$, $OP = OS$, $OQ = OR$
5 $CM = DM$, $\angle OMC = \angle OMD$
6 $OW = OY$, $XW = XY$, $\angle XWO = \angle XYO$

B **1** 8 cm **2** 14 cm **3** 24 cm **4** 18 cm
5 6 cm **6** 8 cm **7** 13 cm **8** 20 cm **9** 70°
10 65° **11** 12 cm **12** 15 cm **13** 24 cm
14 30 cm **15** 10 cm

C **1** **a** 8 cm **b** 11.5 cm **c** 19.5 cm
2 **a** 5 cm **b** 9 cm **c** 14 cm
3 **a** 30 cm **b** 840 cm^2

14—Angle properties of circles

A **1** **d** 90° **2** **d** yes **3** **e** $\angle AOB = 2 \times \angle ACB$

B **1** 90° **2** 30° **3** 25° **4** 90° **5** 34°
6 76° **7** 82° **8** 43° **9** 86° **10** 39°
11 150° **12** 28° **13** 25° **14** 42° **15** 108°
16 27° **17** 78° **18** 120° **19** 40° **20** 36°

C **1** $a = 63°$, $b = 63°$ **2** $a = 58°$, $b = 58°$
3 $a = 104°$, $b = 52°$ **4** $a = 90°$, $b = 90°$
5 $a = 96°$, $b = 132°$ **6** $a = 72°$, $b = 36°$
7 $a = 115°$, $b = 65°$ **8** $a = 96°$, $b = 42°$, $c = 42°$

15—Cyclic quadrilaterals

A **2** 180° **3** 180° **5** yes **7** yes **9** yes
11 yes

B **1** $a = 105°$ **2** $a = 110°$, $b = 95°$ **3** $x = 115°$
4 $x = 95°$ **5** $a = 96°$, $b = 84°$ **6** $x = 100°$
7 $a = 70°$, $b = 75°$ **8** $x = 110°$
9 $a = 98°$, $b = 66°$ **10** $x = 70°$, $y = 110°$
11 $a = 96°$, $b = 102°$ **12** $x = 86°$, $y = 125°$
13 $a = 75°$, $b = 105°$ **14** $x = 102°$, $y = 100°$
15 $a = 108°$, $b = 64°$ **16** $x = 115°$, $y = 122°$

C **1** yes **2** yes **3** no **4** no **5** yes
6 no **7** yes **8** yes

16—Tangents to circles

A **1** $a = 90$ **2** $x = 15$ cm **3** $a = 90$, $b = 90$, $x = 14$ cm
4 $a = 30$, $x = 12$ cm **5** $a = 32$, $x = 11$ cm
6 $a = 27$, $x = 18$ cm **7** $a = 25$, $b = 65$, $c = 65$
8 $a = 60$, $b = 30$, $c = 30$ **9** $a = 90$, $b = 90$, $c = 25$, $d = 25$

B **1** $a = 65$, $b = 65$ **2** $a = 65$, $x = 9$ cm
3 $a = 66$, $b = 66$, $x = 14$ cm **4** $a = 70$, $b = 40$
5 $a = 66$, $b = 48$ **6** $a = 62$, $b = 62$, $x = 17$ cm **7** 15 cm
8 24 cm **9** 8 cm **10** 39 cm **11** 20 cm
12 16 cm

C **1** $x = 27$ **2** $x = 65$ **3** $a = 66$, $b = 66$, $c = 24$
4 $a = 35$, $b = 35$ **5** $x = 10$ cm, $y = 10$ cm
6 $a = 54$, $b = 72$, $c = 36$

17—The angle in the alternate segment

A **4** They are equal. **5** They are equal.

B **1** 55° **2** 50° **3** 115° **4** 68° **5** 62°
6 70° **7** 38° **8** 50° **9** 50° **10** 72°
11 75° **12** 85°

C **1** $x = 50°$, $y = 20°$ **2** $a = 80°$, $b = 30°$
3 $x = 40°$, $y = 40°$ **4** $x = 40°$, $y = 40°$
5 $x = 50°$, $y = 50°$ **6** $x = 35°$, $y = 35°$
7 $x = 50°$, $y = 100°$ **8** $x = 35°$
9 $x = 70°$, $y = 70°$ **10** $x = 55°$, $y = 55°$
11 $a = 80°$, $b = 15°$ **12** $a = 80°$, $b = 75°$, $c = 80°$

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18—Intersecting chords and secants

A **4** yes **5** yes **8** yes

B **1** 8 cm **2** 9 cm **3** 12 cm **4** 16 cm
5 12.5 cm **6** 16 cm **7** 18 cm **8** 4 cm
9 28 cm **10** 6.2 cm

C **1** 8 cm **2** 12 cm **3** 12 cm **4** 10 cm
5 17.7 cm **6** 17.2 cm

19—Review — Properties of circles

A **1** diameter **2** radius **3** chord **4** sector
5 segment **6** arc AB **7** tangent **8** secant
9 Equal *chords* are the same *distance* from the centre of the circle.
10 Equal chords subtend equal *angles* at the centre of the *circle.*
11 The *perpendicular* bisector of a chord passes *through* the *centre* of the circle.
12 Angles at the *circumference* of a circle, *standing* on the same arc, are *equal* in size.
13 The angle subtended by an *arc* at the centre of a circle is twice the size of an angle *subtended* by the same arc at the circumference.
14 The *opposite* angles of a cyclic *quadrilateral* add to 180°.

B **1** $x = 9$ cm **2** $x = 8$ cm **3** $x = 75°$ **4** $x = 15$ cm
5 $x = 90°$ **6** $x = 25°$ **7** $x = 84°$ **8** $x = 43°$
9 $x = 75°$ **10** $x = 110°$ **11** $x = 42°$ **12** $x = 43°$
13 $x = 85°$, $y = 70°$ **14** $x = 92°$, $y = 103°$
15 $x = 80°$ **16** $x = 96°$, $y = 55°$ **17** $x = 90°$
18 $x = 16$ cm **19** $x = 55°$ **20** $x = 68°$

C **1** $a = 46°$, $b = 46°$ **2** $a = 130°$, $b = 65°$
3 $a = 156°$, $b = 102°$ **4** $a = 76°$, $b = 52°$, $c = 52°$
5 $x = 24$ cm **6** $x = 40°$, $y = 25°$
7 $a = 66°$, $y = 66°$ **8** 26.7 cm **9** 31.4 cm
10 56.6 cm **11** 70.7 cm **12** 6 cm **13** 36 cm
14 4 cm **15** 12 cm

20—Circles and deductive reasoning

A **1** 80° (Equal chords subtend equal angles at the centre of the circle.)
2 12 cm (Equal angles, at the centre of a circle, stand on equal chords.)
3 8 cm (Equal chords are the same distance from the centre of the circle.)
4 10 cm (Equal chords are the same distance from the centre of the circle.)
5 90° (The angle in a semicircle is a right angle.)
6 100° (The angle at the centre of a circle is twice the size of an angle subtended by the same arc at the circumference.)
7 65° (Angles at the circumference of a circle, standing on the same arc, are equal in size.)
8 65° (The opposite angles of a cyclic quadrilateral add to 180°.)
9 58° (The angle at the centre of a circle is twice the size of an angle subtended by the same arc at the circumference.)

B **1** $x = 65°$ ($\angle ACB = 90°$, angle in a semicircle, then apply the angle sum of ΔACB) $y = 50°$ (The angle at the centre of a circle is twice the size of the angle subtended by the same arc at the circumference. $\angle BOC = 2 \times \angle BAC$)
2 **a** four (*DOE, EOF, FOG* and *DOG*)
b 65° (*DEFG* is a cyclic quadrilateral. Its opposite angles must add to 180°, $\angle DEF + \angle DGF = 180°$)
3 **a** 40° (Equal chords subtend equal angles at the circumference.)
b $\angle ABD = \angle BDC$
$\therefore$ $AB \parallel DC$ (A pair of alternate angles are equal.)
4 **a** 50° (The angle between a tangent and a chord, drawn to the point of contact, is equal to the angle in the alternate segment.)
b $\angle PCD = 50°$ (base angles of the isosceles triangle, $\angle PCD = \angle PDC$)
$\therefore \angle PCD = \angle BPC$ (both equal 50°)
$\therefore$ $AB \parallel DC$ (A pair of alternate angles are equal)
5 **a** In the Δs *AMO* and *BMO*,
$AM = MB$ (given)
$OA = OB$ (radii of the circle)
OM is common
$\therefore \Delta AMO \equiv \Delta BMO$ (SSS test)

b $\angle AMO = \angle BMO$ (matching angles of the congruent triangles)
$\angle AMO = 90°$ (AB is a straight line)
$\therefore$ $OM \perp AB$

6 **a** In the Δs *AOB* and *DOC*,
$OA = OD$ (radii of the circle)
$\angle AOB = \angle DOC$ (given)
$OB = OC$ (radii of the circle)
$\therefore \Delta AOB \equiv \Delta DOC$ (SAS test)

b $AB = CD$ (matching sides of the congruent triangles)

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ISBN 978-0-99542-759-4